

A conservative, interface-resolved, compressible framework for the modeling and simulation of liquid/gas phase change

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ABSTRACT

This paper presents a method for simulating evaporation in a compressible, interface-resolved framework appropriate for modeling problems of engineering interest. In order to achieve robustness and broad applicability, the method has been designed to discretely enforce consistent mass and thermal energy transport at the phase interface, to globally conserve mass, momentum, and energy, and to be capable of modeling compressible and incompressible systems. Verification is performed via the Sod-shock test, one-dimensional heat conduction, evaporation from a planar interface, and evaporation of three-dimensional droplets. Convergence with increasing mesh resolution is demonstrated in all tested configurations, and conservation is maintained near machine precision for a translating droplet. Conservation and accurate phase change rates are preserved at the low numerical resolutions commonly encountered in engineering calculations. Following verification, the method is validated by comparison to an empirical correlation for evaporating droplets in high temperature crossflow, and the presentation concludes with the simulation of an iso-octane spray at conditions representative of gasoline direct injection. Successful verification, validation, and demonstrated practical utility suggest the method to be an accurate, efficient, and robust approach for the study of phase change in engineering systems.

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1. Introduction

Liquid/gas phase change plays a fundamental role in many important engineering processes, including thermal management [1,2], particle synthesis [3–5], and power generation. Predictive simulations can elucidate the impacts of phase change on these engineering systems, but phase change introduces significant numerical complications to the already challenging task of simulating interfacial flows. In addition to discontinuous fluid properties and surface tension, a liquid/gas interface undergoing phase change is characterized by interdependent discontinuities in velocity, pressure, heat flux, and vapor species mass fraction. As a simplification, modeling efforts have historically focused on nearly spherical droplets with prescriptive sub-models for heat transfer and phase change (i.e. dispersed Lagrangian droplet models) [6,7]. Droplet models are useful when the liquid phase is dispersed and the flow environment is well-understood, but their underlying assumptions are not satisfied outside this regime (however, their application is widespread in all regimes due to low computational costs). The present work focuses on predictive, interface-resolved modeling of phase change in systems with arbitrary, time-dependent interfaces. Modeling phase change in this context requires strategies for (1) transporting the phase interface, (2)

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defining the interfacial fluxes, the mode of phase change, and the rate of phase change, and (3) robustly and conservatively integrating the Navier-Stokes equations in the presence of large, discontinuous changes in fluid properties, as well as discontinuities in fluid velocity at the phase interface.

Numerical methods have been developed to perform interface-resolved simulations of phase change in the context of volume of fluid (VOF) methods [8–14], level set (LS) methods [15–18], and front tracking methods [19–23]. Front tracking methods have the potential to be highly accurate because the interface is tracked directly, but numerical difficulties associated with deformed interfaces and topology change have constrained their application primarily to boiling films [21] and water columns [22]. Interface-capturing methods – based on the VOF or LS – have been applied more broadly because topology changes occur implicitly. The absence of a precisely defined interface, however, introduces the requirement of interface reconstruction, and potential ambiguity in energy and mass transfer at the interface due to phase change. Level set methods are amenable to a ghost fluid method (GFM) treatment of the interfacial jump conditions [17], which allows for accurate prediction of phase change when the surrounding properties are well-defined. Despite the potential for high accuracy, the level set function is not a conserved quantity, and poor conservation behavior has been reported [24] when liquid structures are poorly resolved by the computational mesh – engineering scale applications typically contain a wide range of length scales, and poorly resolved structures are common. Volume of fluid methods, in contrast, conserve mass, but do not provide precise representations of the interface [16].

Regardless of the computational scheme used to identify and transport the interface, correctly defining the interfacial thermodynamic state is a prerequisite for determining the rate of phase change [10,25], and for correctly identifying the relevant phase change mode as boiling, evaporation, or condensation. Boiling characterizes the transition of pure liquid to saturated vapor, initiated within the liquid or on a solid surface [17]. Interface-resolved models designed for the boiling regime generally assume incompressibility and constant density in both phases, and set the interface temperature T_Γ equal to the saturation temperature T_{sat} evaluated at the far-field pressure $T_\Gamma = T_{sat}(p_\infty)$ [8,20,18]. Under these assumptions, the energy transport equation is solved with a Dirichlet temperature applied to the interface. While these assumptions appropriately describe some phase change systems, they are not appropriate for systems with strong spatio-temporal variations in interfacial state. For example, the Dirichlet temperature assumption appropriately describes the heat-transfer-dominated regime of bubble growth, but not the inertia-controlled stage due to the time variation of interface temperature associated with variable bubble pressure [19,25]. Additional complication is introduced by the presence of non-condensable gas species, which allow the liquid to undergo evaporation at interface temperatures below the saturation temperature.

Evaporation describes the spontaneous transition of liquid into vapor below the saturation temperature, where the vapor comprises one component of a gaseous mixture of species (water vapor evaporating into an air/vapor mixture, for example). Evaporation not only depends on the interface temperature, the interfacial heat flux, and the fluid properties, but also on the rate of vapor diffusion away from the interface [15,11,13]; heat and mass transport are not independent at an evaporating interface.

A variety of strategies have addressed the coupled mass and energy transport at evaporating interfaces. Building on a GFM strategy for the Stefan flow problem [26], Tanguy et al. [15] extrapolate the interface temperature from the liquid phase, use the Clausius-Clapeyron relation to compute the interfacial vapor mass fraction (as a function of p_∞), and compute the evaporation rate as a function of the interfacial vapor mass fraction and the rate of vapor mass diffusion. Stability challenges associated with this approach are overcome by solving the species transport equation with a Robin boundary condition, retrieving the interface temperature from the Clausius-Clapeyron relation (assuming p_∞), and then solving the temperature transport equation with a heterogeneous interface temperature [17,24,27]. Ma and Bothe [11] introduce a fixed-point expression to enforce consistency between the interface temperature, the saturation pressure, the rate of phase change, and the rate of interfacial mass diffusion, assuming an error function profile for the sub-grid scale temperature. Palmore and Desjardins [13] also leverage the concept of consistency, but apply the fixed-point iteration directly to a first-order discretization of the interfacial fluxes defined on the geometrically reconstructed interface, thereby achieving a discretely consistent prediction of phase change. These models have been presented with varying levels of verification and success (and many efforts are ongoing), but all of these methods have been designed for the incompressible regime, which limits their applicability to a subset of the relevant problems of interest.

Many multiphase flow applications do not satisfy the incompressible assumption, including the breakup of droplets in a shock tube [28] and the atomization dynamics of certain types of liquid injection (liquid diesel injection at very high-pressure can produce shock waves in the gas phase [29], for example). Moreover, there are fluid models that require the accommodation of large density variations, such as the homogenized fluid model for the liquid-vapor mixture produced by cavitation [30]. Simulating these compressible systems requires numerical methods capable of simultaneously treating compressibility and phase change. Prior work has addressed the compressible phase change problem with a wide range of fidelity. Duret et al. [12] propose a single-fluid, pressure-based method for compressible flows with a constant vaporization source applied directly to the continuity and pressure equations. An approximate Riemann solver has been coupled with a GFM to solve the Euler equations for a boiling 3D drop [31]. Including a larger set of transport mechanisms, Houim and Kuo [16] present a two-dimensional, compressible gas-dynamics methodology based on the GFM that accounts for reaction, vaporization, and cavitation. In contrast to the sharp-interface GFM approaches, four equation models consider mixture equations for transport of mass, momentum, and energy, in addition to a partial density transport equation. These lumped methods are robust and applicable to engineering calculations, but suffer from uncertainty in computing the rate of phase change due to an ambiguous description of the interface [32–35].

This paper presents a computational phase change model for interface-resolved simulations of compressible multiphase flows with non-trivial interface topology. This capability is required, for example, to model the primary atomization dynamics encountered in the direct injection of liquid fuel in a combustion vessel, but its application extends to a much broader domain, including refrigeration and cryogenics. In the interest of clarity and conciseness, the present scope is limited to phase change in the presence of a non-condensable species, at temperatures below the saturation temperature, with a straight-forward extension to boiling reserved for future work. The surveyed literature suggests a predictive phase change model must enforce: (1) a consistent definition of mass diffusion, thermal diffusion, and phase change rate; (2) conservation of mass, momentum, and energy; and (3) an accurate treatment of the normal velocity discontinuity at the interface. In order to satisfy these requirements, the asymptotically preserving methodology of Jemison et al. [36] has been selected to provide the foundation of the development efforts. The asymptotically preserving scheme recovers an incompressible pressure projection as the sound speed approaches infinity, and allows for the simulation of flows with widely varying Mach numbers and fluid properties, without the computational burden of resolving the acoustic wave speed [37,36]. Because phase change introduces unique couplings between mass, momentum, and energy transport, modifications to the underlying numerical schemes are required. An outline for incorporating phase change into the framework of Jemison et al. [36] was first presented by Wenzel and Arienti [38]; the focus of the present effort is to complete the development, verification, and validation of this methodology.

The paper begins with a presentation of the governing equations and the interfacial conditions, then presents the numerical methods and their verification, and concludes with validation and a capability demonstration. The relevant multiphase governing equations and interfacial boundary conditions are defined in Sections 2 and 3, respectively. The governing equations and interfacial conditions are then re-cast in a semi-discrete form to introduce the operator-split time integration strategy of Jemison et al. [36]. The first-order time integration is shown to be comprised of sequentially applied sub-operations corresponding to advection, diffusion, and pressure effects. Incorporating phase change requires modifications to each sub-operation, including the introduction of sharp-interface diffusion. For reference in navigating the contributions of this manuscript, the following developments appear in Section 4:

- Modifications to the time integration scheme to facilitate the incorporation of phase change;
- Introduction of phase-specific advection velocities into the cell-integrated, semi-Lagrangian advection scheme;
- A modified procedure for defining face-centered velocities at the post-advection state;
- A sharp-interface diffusion strategy, leveraging the geometrically reconstructed interface defined by the moment of fluid (MOF) method [39] (inspired by prior work of Palmore and Desjardins [13] and Ma and Bothe [11], the discretization ensures discrete consistency between the rate of phase change and the rate of mass and energy transport at the interface);
- A mixed-time integration scheme for diffusion and phase change that eliminates the stiff and computationally expensive implicit system typically associated with sharp-interface discretizations;
- A modified pressure projection for the single-fluid velocity that accounts for the velocity divergence of Stefan flow, along with a procedure for computing the corresponding phase-specific velocities;
- Auxiliary procedures required to achieve conservation and robustness.

Verification of the resulting numerical method is presented in Section 5. The Sod-shock tube problem is used to verify compressible single phase performance relative to the foundational method of Jemison et al. [36]. Multiphase diffusion and evaporation are then evaluated in test problems of incrementally increasing complexity. The mixed-time integration strategy and sharp-interface diffusion discretization are verified via transient heat transfer between semi-infinite domains of water and air. Phase change is then incorporated by considering evaporation from a planar interface with a fixed interfacial vapor mass fraction. The full coupling of interfacial transport is examined via the evaporation of three-dimensional droplets, where the evaporation rate is evaluated and conservation is verified. After verification, validation is presented for evaporating droplets in crossflow at high mass transfer rates. Finally, applicability to relevant engineering problems is demonstrated via simulation of high-pressure liquid iso-octane injection with real fluid thermodynamics.

2. Governing equations

We consider a compressible liquid/gas system, where the gas phase is a mixture of a non-condensable species and the gaseous vapor produced by the liquid. The convention observed throughout this work defines “phase” as the state of matter (liquid or gas), while “species” indicates the chemical composition (non-condensable gas or vapor). When referring to fluid or geometric properties, a subscript index m identifies the relevant phase, where $m = l$ in the liquid phase and $m = g$ in the gas phase. The expression η_m therefore indicates a generic property without specification of phase, whereas η_l and η_g indicate the liquid and gas properties, respectively. Quantities defined as a volume-average or mass-average across all present phases appear without a subscript. The governing equations used in this work require a mathematical definition for the spatial distribution of phase, as well as a definition for volume-averaged quantities (applicable to interfacial computational cells).

The presence of phase m at position $\mathbf{x} = (x, y, z)$ at time t is defined as true (phase m is located at position $\mathbf{x}$ at time t) or false (phase m is not located at position $\mathbf{x}$ at time t) by an indicator function $\chi_m(t, \mathbf{x})$,

$$\chi_m(t, \mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in m; \\ 0, & \mathbf{x} \notin m. \end{cases} \quad (1)$$

The volume fraction of phase m within a region Ω of volume V is defined as a function of the phase indicator according to

$$\phi_m = \frac{1}{V} \int_{\Omega} \chi_m d\Omega. \quad (2)$$

Similarly, the volume-averaged density ρ corresponding to Ω is computed from the product of phase indicator χ_m and density ρ_m ,

$$\rho = \sum_m \frac{1}{V} \int_{\Omega} \rho_m \chi_m d\Omega, \quad (3)$$

where summation is performed over phases m . The density-weighted average velocity $\mathbf{u}$, commonly called the “single-fluid velocity”, is computed as a function of the volume-averaged density ρ , the phase-specific density ρ_m , the phase-specific velocity $\mathbf{u}_m$, and the phase-specific phase indicator χ_m , according to

$$\mathbf{u} = \frac{1}{\rho} \sum_m \frac{1}{V} \int_{\Omega} \rho_m \mathbf{u}_m \chi_m d\Omega. \quad (4)$$

Multiplying the volume-averaged density and single-fluid velocity returns the single-fluid momentum,

$$\rho \mathbf{u} = \sum_m \frac{1}{V} \int_{\Omega} \rho_m \mathbf{u}_m \chi_m d\Omega. \quad (5)$$

Leveraging the single-fluid quantities defined by Eqs. (1)-(5) allows the Navier-Stokes equations to be expressed in the form of System (6), where the general notation has been inspired by [36].

$$\frac{\partial}{\partial t} \begin{pmatrix} \phi_m \\ \rho_m \phi_m \\ \rho \mathbf{u} \\ \rho_g Y_g \phi_g \\ \rho_m E_m \phi_m \end{pmatrix} + \nabla \cdot \begin{pmatrix} \phi_m \mathbf{u}_m \\ \rho_m \phi_m \mathbf{u}_m \\ \sum_m [\mathbf{u}_m \otimes \rho_m \phi_m \mathbf{u}_m] + p \mathbf{I} - \tau \\ [\rho_g Y_g \mathbf{u}_g - \mathbf{q}_Y] \phi_g \\ [\rho_m E_m \mathbf{u}_m + p \mathbf{u} - \mathbf{u} \cdot \tau - \mathbf{q}_m] \phi_m \end{pmatrix} = \begin{pmatrix} \phi_m \nabla \cdot \mathbf{u}_m \\ 0 \\ \sigma \kappa \mathbf{n} \delta_{\Gamma} \\ 0 \\ 0 \end{pmatrix} \quad (6)$$

The compressible flow system with phase change is modeled by the formulation of the Navier-Stokes equations defined by System (6). The first equation (top row) defines transport of the volume fraction of phase m , ϕ_m , by the phase-specific velocity $\mathbf{u}_m$. Conservation of mass for each phase m appears as the second equation. The third equation defines conservation of single-fluid momentum, where ρ is the density, p is the pressure, $\mathbf{I}$ is the identity tensor, and τ is the viscous stress tensor,

$$\tau = \mu \left[(\nabla \mathbf{u} + [\nabla \mathbf{u}]^T) - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right], \quad (7)$$

which is a function of the dynamic viscosity μ and the single-fluid velocity $\mathbf{u}$. The final term in the momentum equation, appearing on the right-hand side, corresponds to surface tension, which is proportional to the surface tension coefficient σ and the mean interfacial curvature κ , and acts along the interface normal vector $\mathbf{n}$ (where $\delta_{\Gamma} = 1$ on the interface, and $\delta_{\Gamma} = 0$ elsewhere). The viscous and pressure terms in the single-fluid momentum equation are averaged following the procedures of [36]. Nonlinear advective transport of momentum is performed for each phase as a function of phase-specific properties, and the advective contribution to the single-fluid momentum is defined by summation over all phases. The fourth equation defines species transport in the multi-species gas phase as a function of Fick's law,

$$\mathbf{q}_Y = -\rho_g \mathcal{D}_g \nabla Y_g, \quad (8)$$

where Y_g is the vapor mass fraction and $\mathcal{D}_g$ is the assumed binary coefficient of mass diffusion. The fifth and final equation defines conservation of total energy E , comprised of internal and kinetic energies,

$$E_m = e_m + \frac{1}{2} \mathbf{u} \cdot \mathbf{u}. \quad (9)$$

In the energy equation, thermal diffusion $\mathbf{q}_m$ encompasses the effects of heat conduction $\mathbf{q}_{T,m}$ and enthalpy transport due to mass diffusion $\mathbf{q}_{h,m}$, such that

$$\mathbf{q}_m = \mathbf{q}_{T,m} + \mathbf{q}_{h,m}. \quad (10)$$

Heat conduction is modeled by Fourier's law,

$$\mathbf{q}_{T,m} = -k_m \nabla T_m, \quad (11)$$

where $\mathbf{q}_{T,m}$ is the conductive heat flux vector and k_m is the thermal conductivity. Consistent with the species transport equation, enthalpy transport in the gas phase due to mass diffusion is modeled by Fick's law,

$$\mathbf{q}_{h,m} = \begin{cases} 0 & \text{if } m = l \\ (h_v - h_{nc}) \mathbf{q}_Y & \text{if } m = g \end{cases} \quad (12)$$

where h_v is the enthalpy of the vapor and h_{nc} is the enthalpy of the non-condensable gas (this expression has been simplified for the two-species gas considered in this work).

The liquid and gas phases governed by System (6) interact via interface conditions for mass, momentum, and energy transport between the liquid and the gas by means of thermal diffusion, phase change, and surface forces. Solving System (6) subject to this set of interface conditions (in addition to relevant initial and boundary conditions) defines the spatio-temporal evolution of the multiphase, multispecies, compressible flow system undergoing phase change.

3. Interface conditions

The interface between the liquid and gas phase is a surface Γ . The difference in fluid properties evaluated on the liquid and gas sides of the interface is expressed with bracket notation, where the discontinuity in property η at interfacial point $\mathbf{x}_\Gamma \in \Gamma$ is given by

$$[\eta(\mathbf{x}_\Gamma)] = \eta_g(\mathbf{x}_\Gamma) - \eta_l(\mathbf{x}_\Gamma). \quad (13)$$

Subscripts g and l denote the gas and liquid sides of the interface, respectively. In this section, interfacial conditions are specified in interface-fitted coordinates, characterized by an interface unit normal vector $\mathbf{n}$ (pointing from the liquid toward the gas), and the corresponding tangent vector $\mathbf{t}$. We assume a subsonic, non-characteristic interface [40], neglecting kinetic energy transfer in the definition of jump conditions. Consistent with a single-fluid momentum equation, discontinuities in tangential velocity are also neglected.

The area-specific rate of phase change $\dot{m}$ is coupled to thermal transport across the interface by

$$\hat{H}\dot{m} = [-\mathbf{q}_T] \cdot \mathbf{n}, \quad (14)$$

where $\hat{H}$ is the enthalpy of phase change [17,16,13]. Evaporation and boiling correspond to $\dot{m} > 0$, while $\dot{m} < 0$ for condensation. Assuming local thermodynamic equilibrium corresponding to $[T] = 0$, the mole fraction of vapor at the interface X_Γ is equal to the ratio of the vapor saturation pressure p_s (evaluated at the interfacial temperature T_Γ), to the pressure of the gas at the interface [11],

$$X_\Gamma = \frac{p_s}{p_g} \Big|_\Gamma. \quad (15)$$

The vapor mole fraction and vapor mass fraction Y_Γ are related by

$$Y_\Gamma = \frac{X_\Gamma W_v}{X_\Gamma W_v + (1 - X_\Gamma) W_{nc}}, \quad (16)$$

where W_v and W_{nc} are the molecular weights of the vapor and non-condensable gas, respectively.

When boiling, the interfacial saturation pressure and gas pressure are equal, $p_s = p_g(T_\Gamma, \rho_\Gamma)$. When the gas pressure exceeds the saturation pressure, the liquid cannot boil, and evaporates instead. During the evaporation of a single-component liquid, the rate of phase change $\dot{m}$ is coupled to the rate of vapor diffusion away from the interface by

$$\dot{m} = \frac{-\mathbf{n} \cdot \mathbf{q}_Y \Big|_\Gamma}{Y_\Gamma - 1}, \quad (17)$$

where Y_Γ is the vapor mass fraction at the interface, defined by Eq. (16) [15,11,13]. Evaporation is therefore a coupled phenomenon, where the evaporation process simultaneously satisfies Eqs. (14)-(17) [13]. Conversely, the boiling rate is controlled entirely by the interfacial energy balance of Eq. (14). Regardless of the mechanism of phase change, mass transport between the liquid and gas phases generates a momentum response at the interface.

As liquid transitions to vapor, it expands from the local liquid density to the local gas density, inducing a gas phase velocity in the interface-normal direction called Stefan flow. The discontinuity in normal velocity is related to the fluid densities and the rate of phase change by

$$[u_n] = \left[\frac{1}{\rho} \right] \dot{m}. \quad (18)$$

This phase-change-induced velocity couples with surface tension to produce a discontinuity in pressure across the interface

$$[p] = -\kappa\sigma - \dot{m}[u_n], \quad (19)$$

where κ is the interfacial curvature and σ is the surface tension coefficient. Additional conditions associated with the neglected tangential stresses discontinuities are outlined by Houim and Kuo [16] for compressible flows, but are neglected in this work. Substitution of Eq. (18) into Eq. (19) shows the pressure jump associated with phase change to be negative definite for $\rho_l > \rho_g$.

4. Numerical methods

The numerical methodology proposed for solving the governing equations presented in Section 2, subject to the interface conditions presented in Section 3, appears in a six-part progression: (1) the temporal discretization, outlining the operator-split time integration; (2) the spatial representation of phase and state via the moment of fluid method; (3) advection of cell-centered and face-centered quantities; (4) finite-volume discretization of diffusion terms and phase change; (5) pressure correction; and (6) auxiliary operations.

4.1. Temporal discretization

This section outlines the operator-split time integration scheme in a semi-discrete format, with details regarding spatial discretization reserved for subsequent sections. The time-evolving quantities of interest are contained in the state vector $\mathbf{Q}$,

$$\mathbf{Q} = \begin{pmatrix} \phi_m \\ \rho_m \phi_m \\ \rho \mathbf{u} \\ \rho_g Y_g \phi_g \\ \rho_m E_m \phi_m \end{pmatrix}. \quad (20)$$

The Navier-Stokes equations (6), subject to the interfacial conditions defined in Section 3, are decomposed into three vectors accounting for different sources of temporal change in $\mathbf{Q}$: the advection vector $\mathbf{A}$; the diffusion and phase change vector $\mathbf{D}$; and the pressure vector $\mathbf{P}$. This decomposition allows the temporal rate-of-change of the state vector $\mathbf{Q}$ to be expressed by

$$\frac{\partial \mathbf{Q}}{\partial t} + \mathbf{A} + \mathbf{D} + \mathbf{P} = 0, \quad (21)$$

where

$$\mathbf{A} = \begin{pmatrix} \nabla \cdot (\phi_m \mathbf{u}_m) - \phi_m \nabla \cdot \mathbf{u}_m \\ \nabla \cdot (\rho_m \phi_m \mathbf{u}_m) \\ \sum_m \nabla \cdot (\mathbf{u}_m \otimes \rho_m \phi_m \mathbf{u}_m) \\ \nabla \cdot (\rho_g Y_g \phi_g \mathbf{u}_g) \\ \nabla \cdot (\rho_m E_m \phi_m \mathbf{u}_m) \end{pmatrix}; \quad \mathbf{D} = \begin{pmatrix} -\dot{\phi}_m \\ -\dot{\mathcal{M}}_m \\ -\nabla \cdot \boldsymbol{\tau} \\ \nabla \cdot (\rho_g \mathcal{D}_g \phi_g \nabla Y_g) - \dot{\mathcal{Y}} \\ -\nabla \cdot (\mathbf{u} \cdot \boldsymbol{\tau} + \mathbf{q}_m) \phi_m - \dot{\mathcal{E}}_m \end{pmatrix}; \quad \mathbf{P} = \begin{pmatrix} 0 \\ 0 \\ \nabla p \mathbf{I} \\ 0 \\ \nabla \cdot (p \mathbf{u} \phi_m) \end{pmatrix}. \quad (22)$$

The diffusion and phase change vector $\mathbf{D}$ contains a number of terms that do not explicitly appear in Eq. (6), namely, $\dot{\phi}_m$, $\dot{\mathcal{M}}_m$, $\dot{\mathcal{Y}}$, and $\dot{\mathcal{E}}_m$. These terms contain the temporal contribution to $\mathbf{Q}$ attributable to thermal exchange at the liquid/gas interface, including the effects of phase change. These terms do not appear in Eq. (6) because they are boundary conditions applied at the liquid/gas interface, and therefore are not rigorously part of the governing equations. However, as these terms are the focus of the present work, they have been explicitly included in Eqs. (21)–(22) to maximize clarity regarding their application. The effects of surface tension have been absorbed into the pressure in Eq. (22) because surface tension is computationally applied via the ghost fluid method.

The state vector $\mathbf{Q}$ is advanced in time through the operator split approach schematically outlined in Fig. 1. Panel (a) illustrates the temporally first-order accurate time-stepping structure. The first and last blocks in panel (a) define the simulation starting and stopping logic, and the outer-most loop defines the advancement of the algorithm in time by increments of Δt . The interior gray box contains sequential operations that increment the state vector from time level n , denoted by $\mathbf{Q}^n$, to the time-advanced state, denoted by $\mathbf{Q}^{n+1}$.

Panel (b) of Fig. 1 outlines the operator-split integration procedure introduced by Jemison et al. [36], which has been expanded to include thermal diffusion and phase change in the diffusion portion of the operator-split procedure. Integrating the quantities of interest $\mathbf{Q}$ from time level n to time level $n+1$ is accomplished by first performing advection

$$\mathbf{Q}^a = \mathbf{Q}^n + \mathbf{A}^n \Delta t, \quad (23)$$

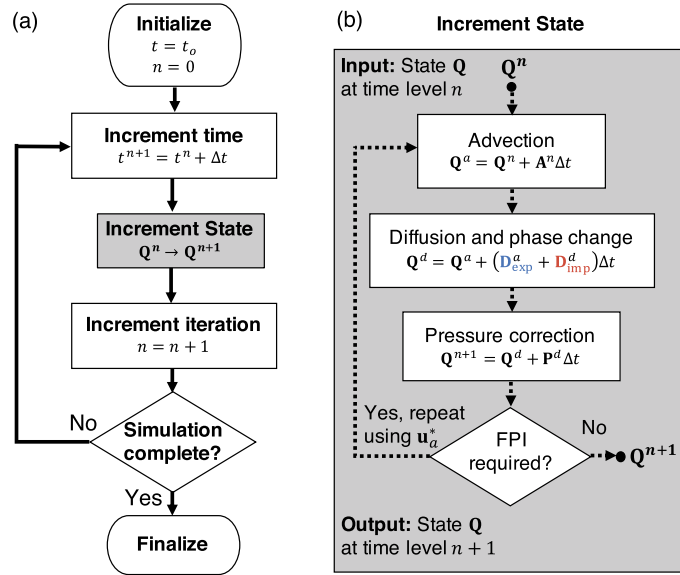


Fig. 1. Flow chart of the operator-split integration strategy, illustrating the sequential implementation of the advection operator $\mathbf{A}^n$, the explicit diffusion operator $\mathbf{D}_{exp}^a$ (blue), the implicit diffusion and phase change operator $\mathbf{D}_{imp}^d$ (red), and the pressure operator $\mathbf{P}^d$. Sequential operation advances the state vector $\mathbf{Q}$ from time level n to time level $n + 1$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

where $\mathbf{A}^n \Delta t$ defines the change in $\mathbf{Q}$ attributable to advection, evaluated explicitly (via first-order forward-Euler integration) at time level n . Superscript a indicates the intermediate time level corresponding to the post-advection state. Advection is performed as a function of the face-centered advective velocity, $\mathbf{u}_a$. The advective velocity and the advection operation defined by Eq. (23) are the focus of Section 4.3. After advection, diffusion and phase change advance the post-advection time level a to the post-diffusion time level d according to

$$\mathbf{Q}^d = \mathbf{Q}^a + (\mathbf{D}_{exp}^a + \mathbf{D}_{imp}^d) \Delta t, \quad (24)$$

where $\mathbf{D}_{exp}^a$ and $\mathbf{D}_{imp}^d$ impose phase change and diffusion effects in a temporally-mixed manner: contributions contained in $\mathbf{D}_{exp}^a$ are discretized explicitly at time level a (corresponding to a first-order forward-Euler integration), while those contained in $\mathbf{D}_{imp}^d$ are implicitly discretized at time level d (corresponding to a first-order backward-Euler integration). The definitions of $\mathbf{D}_{exp}^a$ and $\mathbf{D}_{imp}^d$ are the subject of Section 4.4. In the final operation, a pressure correction applies the effects of pressure

$$\mathbf{Q}^{n+1} = \mathbf{Q}^d + \mathbf{P}^d \Delta t. \quad (25)$$

The pressure correction procedure defined by $\mathbf{P}^d \Delta t$ is the subject of Section 4.5.

Performing the operations defined by Eqs. (23)–(25) constitutes a complete time advancement of the state vector by time increment Δt . However, consistent with Jemison et al. [36], a single fixed point iteration (FPI) is executed for the advective velocity $\mathbf{u}_a$, as indicated by the logical operation following pressure correction in Fig. 1 panel (b). The FPI is performed by executing a first pass through the time-integration procedure as a function of $\mathbf{Q}^n$ and $\mathbf{u}_a^n$, producing an estimate of the time-advanced advective velocity $\mathbf{u}_a^*$. The time-integration is then repeated as a function of $\mathbf{Q}^n$, but with the predicted advective velocity $\mathbf{u}_a^*$. Completing the FPI returns the time-advanced state $\mathbf{Q}^{n+1}$.

4.2. Representation of phase

All operations involved in the time-advancement are sensitive to the interface geometry. The moment of fluid (MOF) method defines the spatial distribution of phase on the computational mesh. Computational cells containing both phases are defined as interfacial cells. In every interfacial cell i , the interface is represented by a plane with normal vector $\mathbf{n}$ and intercept b ,

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_i) + b = 0, \quad (26)$$

where $\mathbf{x}_i$ is the center of cell i . The reconstruction operation produces a planar interface in every interfacial cell i , characteristic of the piecewise linear interface calculation (PLIC) approach. The reconstruction is constrained by Eq. (27)

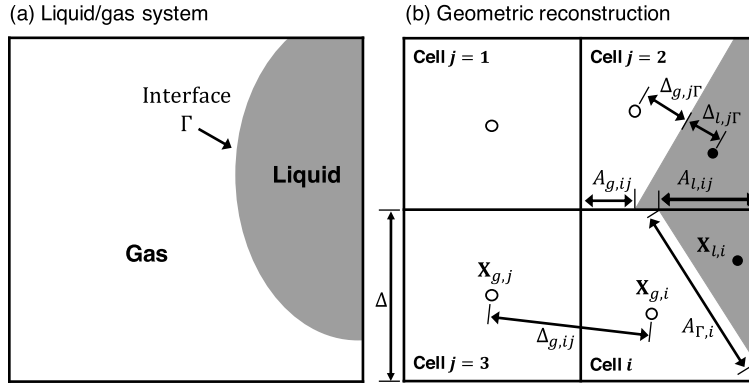


Fig. 2. A hypothetical liquid/gas system, shown in panel (a), is geometrically reconstructed in panel (b). Reconstruction is performed in cell i and neighboring cells, $j = 1, 2$ and 3 . Important geometric quantities resulting from the reconstruction have been identified in panel (b).

$$(\mathbf{n}, b) = \underset{\mathbf{n}, b, \phi_{\text{Ref}} = \phi_{\text{Recon}}}{\text{argmin}} \quad \|\mathbf{X}_{\text{Ref}} - \mathbf{X}_{\text{Recon}}\|, \quad (27)$$

which enforces conservation of mass, and minimizes the deviation between the reconstructed center of mass of the majority phase $\mathbf{X}_{\text{Recon}}$ and the reference center of mass $\mathbf{X}_{\text{Ref}}$. Evolving the distribution of phase in time involves transporting the reference center of mass as a function of the advective velocity, followed by reconstructing the interface in every interfacial cell according to Eq. (27). For details regarding the center of mass transport procedure and the reconstruction minimization problem, the reader is directed to Section 4.1.1 of Jemison et al. [39]; the MOF implementation used in this work is identical, and a repeated description is beyond the present scope.

4.2.1. Geometric quantities

A hypothetical liquid/gas system and the geometric reconstruction via the MOF method are illustrated in Fig. 2. Panel (a) shows the liquid/gas system with interface Γ , and panel (b) shows the reconstructed interface, including geometric quantities relevant to computing interfacial transport rates. The discretization includes four computational cells, each with face length Δ . Interfacial cell i , shown in the lower-right of panel (b), contains gas with a center of mass $\mathbf{X}_{g,i}$, and liquid with a center of mass $\mathbf{X}_{l,i}$. The interfacial area within cell i , $A_{\Gamma,i}$, is computed from the tessellated PLIC reconstruction. Similarly, the distance separating the gas phase center of mass from the interface, referred to as the gas phase interfacial distance $\Delta_{g,j\Gamma}$, is computed from the reconstruction. This interfacial distance is illustrated in cell $j = 2$, along with the analogous quantity for the liquid, $\Delta_{l,j\Gamma}$.

The distance between the gas phase center of mass in cell i and its neighbor cell j is defined by $\Delta_{g,ij} = \|\mathbf{X}_{g,j} - \mathbf{X}_{g,i}\|$, illustrated in Fig. 2 panel (b) between cell i and cell $j = 3$. The distance separating liquid phase centers of mass is similarly defined by $\Delta_{l,ij} = \|\mathbf{X}_{l,j} - \mathbf{X}_{l,i}\|$ (although, not illustrated in the figure for the purpose of visual clarity). The gas phase wetted area of the cell face joining cell i and cell j is defined by $A_{g,ij}$ (with the analogous quantity $A_{l,ij}$ in the liquid). The wetted areas for each phase are taken to be the lesser of the values computed in cells i and j , following the convention of Palmore and Desjardins [13].

4.2.2. Averaged quantities

Fluid properties for each phase are stored at the corresponding center of mass in each cell, constituting a cell-averaged quantity for each phase. The volume-averaged density ρ_i corresponding to cell i is computed from the volume fractions $\phi_{m,i}$ and densities $\rho_{m,i}$ according to

$$\rho_i = \rho_{l,i}\phi_{l,i} + \rho_{g,i}\phi_{g,i}. \quad (28)$$

The density-weighted, single-fluid velocity $\mathbf{u}_i$ is computed as a function of the volume-averaged density from

$$\mathbf{u}_i = \frac{1}{\rho_i} (\rho_{l,i}\mathbf{u}_{l,i}\phi_{l,i} + \rho_{g,i}\mathbf{u}_{g,i}\phi_{g,i}). \quad (29)$$

Equations (28) and (29) can be considered templates for volume-averaging and mass-averaging, respectively.

4.3. Advection

For each computational cell i , quantities of interest $\mathbf{Q}_i$ and centers of mass $\mathbf{X}_i$ are advected by the face-centered advective velocity $\mathbf{u}_a = (u_a, v_a, w_a)$ according to the multiphase, cell-integrated, semi-Lagrangian (CISL) framework presented by Jemison et al. [36]. This directionally-split advection procedure alternates backward tracing, Eulerian Implicit (EI) and forward tracing, Lagrangian Explicit (LE) operations [36]. The pre-advection state advances to the post-advection state through

Algorithm 1 Cell-integrated semi-Lagrangian advection.

```

1: procedure CISL
2:    $\mathbf{Q}^{a1/6} \leftarrow \text{EI}(x, \mathbf{u}_a, \mathbf{Q}^n)$  ▷ Call EI to advect  $\mathbf{Q}^n$  in x-direction with  $u_a$ 
3:    $\mathbf{Q}^{a2/6} \leftarrow \text{LE}(y, \mathbf{u}_a, \mathbf{Q}^{a1/6})$  ▷ Call LE to advect  $\mathbf{Q}^{a1/6}$  in y-direction with  $v_a$ 
4:    $\mathbf{Q}^{a3/6} \leftarrow \text{EI}(z, \mathbf{u}_a, \mathbf{Q}^{a2/6})$  ▷ Call EI to advect  $\mathbf{Q}^{a2/6}$  in z-direction with  $w_a$ 
5:    $\mathbf{Q}^{a4/6} \leftarrow \text{LE}(z, \mathbf{u}_a, \mathbf{Q}^{a3/6})$  ▷ Call LE to advect  $\mathbf{Q}^{a3/6}$  in z-direction with  $w_a$ 
6:    $\mathbf{Q}^{a5/6} \leftarrow \text{EI}(y, \mathbf{u}_a, \mathbf{Q}^{a4/6})$  ▷ Call EI to advect  $\mathbf{Q}^{a4/6}$  in y-direction with  $v_a$ 
7:    $\mathbf{Q}^a \leftarrow \text{LE}(x, \mathbf{u}_a, \mathbf{Q}^{a5/6})$  ▷ Call LE to advect  $\mathbf{Q}^{a5/6}$  in x-direction with  $u_a$ 
8: end procedure

9: function EI( $dir, \mathbf{u}_a, \mathbf{Q}$ )
10:   $\mathbf{Q}^{new} \leftarrow \text{ADVTECTEI}(dir, \mathbf{u}_a, \mathbf{Q})$  ▷ Perform EI advection, then return new state
11:  return  $\mathbf{Q}^{new}$ 
12: end function

13: function LE( $dir, \mathbf{u}_a, \mathbf{Q}$ )
14:   $\mathbf{Q}^{new} \leftarrow \text{ADVTECTLE}(dir, \mathbf{u}_a, \mathbf{Q})$  ▷ Perform LE advection, then return new state
15:  return  $\mathbf{Q}^{new}$ 
16: end function

```

a sequence of six sub-operations. Using superscript $a_{s/6}$ to denote the state after completing the s^{th} sub-operation, the quantities of interest are advanced from the pre-advection state $\mathbf{Q}^n$ to the post-advection state $\mathbf{Q}^a$ according to

$$\mathbf{Q}^n \xrightarrow{\text{El}_x} \mathbf{Q}^{a1/6} \xrightarrow{\text{LE}_y} \mathbf{Q}^{a2/6} \xrightarrow{\text{El}_z} \mathbf{Q}^{a3/6} \xrightarrow{\text{LE}_z} \mathbf{Q}^{a4/6} \xrightarrow{\text{El}_y} \mathbf{Q}^{a5/6} \xrightarrow{\text{LE}_x} \mathbf{Q}^a. \quad (30)$$

The notation El_x denotes application of the Eulerian Implicit operation in the x -direction as a function of the advective velocity u_a , and LE_y the Lagrangian Explicit operation in the y -direction as a function of advective velocity v_a (with the other operations similarly defined). The quantities of interest $\mathbf{Q}_i^n$ and the centers of mass $\mathbf{X}_i^n$ change with each advection operation applied in Eq. (30), while the advective velocity $\mathbf{u}_a$ remains fixed at the initial state (i.e., the same advection velocity is used for each sub-operation). Algorithm 1 provides a pseudo-code description of the CISL procedure. The six steps defined by Eq. (30) are performed over lines 2-7, and leverage functions EI (defined on lines 9-12) and LE (defined on lines 13-16). These functions are shown as “black boxes”, appearing simply to illustrate the dependence on advection direction dir , the advective velocity $\mathbf{u}_a$, and the current state $\mathbf{Q}$. Prior to each advection operation, conserved quantities are reconstructed in single-phase cells with a MINMOD-limited piecewise linear reconstruction, while a piecewise constant reconstruction is applied to each phase in interfacial cells [36]. This reconstruction procedure results in second-order spatial accuracy away from phase interfaces, and first-order spatial accuracy near phase interfaces. In practice, slightly lower than first-order mesh convergence has been demonstrated for this discretization scheme in the context of a variety of flows with shocks [36].

For a complete description of the CISL advection scheme, the reader is directed to the detailed coverage provided by Jemison et al. [36]. The present section instead focuses on two modifications that have been applied to the multiphase CISL method to account for interfacial velocity discontinuities associated with phase change. Here, the advective velocity is uniquely defined for each phase, rather than being commonly defined for both phases, and the procedure for updating the face-centered velocity is altered to preserve interfacial velocity discontinuities. This section defines the advective velocity, introduces the cell-centered and face-centered quantities relevant to the advection procedure, and describes the procedure for updating the face-centered velocity to the post-advection time level. Portions of the discussion are limited to transport in the x -direction – the directionally-split nature of the algorithm allows for straight-forward generalization to three dimensions [36].

4.3.1. CISL with phase-specific advective velocity

In the original multiphase CISL implementation, the face-centered, single-fluid velocity provides the advective velocity for both phases. This definition spuriously generates interfacial area when the velocity is discontinuous at the phase interface. We therefore consider phase-specific advective velocities $\mathbf{u}_{a,l}$ for the liquid phase and $\mathbf{u}_{a,g}$ for the gas phase. This section illustrates how the phase-specific velocities are used to perform advection in the CISL context, while Section 4.5.1 introduces the method for computing the phase-specific advective velocities. Away from the interface, the present advection implementation is identical to the original implementation of Jemison et al. [36] and described by Algorithm 1. Near the interface, the following strategy is applied to develop a CISL approach for phase-specific advective velocities: (1) advect the state vector and moment of fluid variables for each phase with the corresponding phase-specific advective velocity; (2) define the post-advection conserved quantities as the result of the corresponding phase-specific advective velocity operation; (3) define the post-advection spatial distribution of phase as the result of the liquid phase advective velocity operation. Applying these modifications to the CISL procedure outlined in Algorithm 1 results in a new CISL procedure provided in Algorithm 2.

The details of performing CISL advection with phase-specific advective velocities are illustrated by considering the application of Algorithm 2 to the first advection sub-operation, $\mathbf{Q}^n \xrightarrow{\text{El}_x} \mathbf{Q}^{a1/6}$:

Algorithm 2 CISL with phase-specific advective velocities.

```

1: procedure PHASESPECIFICCISL
2:    $\mathbf{Q}_g^{a_{1/6}} \leftarrow \text{El}(x, \mathbf{u}_{a,g}, \mathbf{Q}^a)$                                 ▷ El advection of  $\mathbf{Q}^a$  with gas advective velocity
3:    $\mathbf{Q}_l^{a_{1/6}} \leftarrow \text{El}(x, \mathbf{u}_{a,l}, \mathbf{Q}^a)$                                 ▷ El advection of  $\mathbf{Q}^a$  with liquid advective velocity
4:    $\mathbf{Q}^{a_{1/6}} \leftarrow \text{RECONSTRUCTQ}(\mathbf{Q}_g^{a_{1/6}}, \mathbf{Q}_l^{a_{1/6}})$                 ▷ Reconstruct  $\mathbf{Q}^{a_{1/6}}$  from  $\mathbf{Q}_g^{a_{1/6}}$  and  $\mathbf{Q}_l^{a_{1/6}}$ 
5:    $\vdots$ 
6:    $\mathbf{Q}_g^a \leftarrow \text{LE}(x, \mathbf{u}_{a,g}, \mathbf{Q}^{5/6})$                                 ▷ Loop through y- and z-directions
7:    $\mathbf{Q}_l^a \leftarrow \text{LE}(x, \mathbf{u}_{a,l}, \mathbf{Q}^{5/6})$                                 ▷ LE advection of  $\mathbf{Q}^{5/6}$  with gas advective velocity
8:    $\mathbf{Q}^a \leftarrow \text{RECONSTRUCTQ}(\mathbf{Q}_g^a, \mathbf{Q}_l^a)$                             ▷ LE advection of  $\mathbf{Q}^{5/6}$  with liquid advective velocity
9: end procedure                                                    ▷ Reconstruct  $\mathbf{Q}^a$  from  $\mathbf{Q}_g^a$  and  $\mathbf{Q}_l^a$ 

10: function RECONSTRUCTQ( $\mathbf{Q}_g, \mathbf{Q}_l$ )
11:    $\mathbf{u}_l \leftarrow \mathbf{Q}_l[\phi_l \rho_l \mathbf{u}_l] / \mathbf{Q}_l[\phi_l \rho_l]$ 
12:    $E_l \leftarrow \mathbf{Q}_l[\phi_l \rho_l E_l] / \mathbf{Q}_l[\phi_l \rho_l]$ 
13:    $\mathbf{u}_g \leftarrow \mathbf{Q}_g[\phi_g \rho_g \mathbf{u}_g] / \mathbf{Q}_g[\phi_g \rho_g]$ 
14:    $E_g \leftarrow \mathbf{Q}_g[\phi_g \rho_g E_g] / \mathbf{Q}_g[\phi_g \rho_g]$ 
15:    $Y_g \leftarrow \mathbf{Q}_g[\phi_g \rho_g Y_g] / \mathbf{Q}_g[\phi_g \rho_g]$ 
16:    $\epsilon \leftarrow 10^{-10}$                                                 ▷ Tolerance variable
17:   if  $\mathbf{Q}_l[\phi_g] < \epsilon$  and  $\mathbf{Q}_g[\phi_g] > \epsilon$  then                        ▷ Inconsistent State #1
18:      $\phi_l \leftarrow \mathbf{Q}_g[\phi_l]$ 
19:      $\rho_l \leftarrow \mathbf{Q}_l[\phi_l \rho_l] / \mathbf{Q}_g[\phi_l]$                         ▷ Remedy: define volumes using  $\mathbf{Q}_g$ , rather than  $\mathbf{Q}_l$ 
20:      $\phi_g \leftarrow \mathbf{Q}_g[\phi_g]$ 
21:      $\rho_g \leftarrow \mathbf{Q}_g[\phi_g \rho_g] / \mathbf{Q}_g[\phi_g]$ 
22:   else if  $\mathbf{Q}_l[\phi_g] > \epsilon$  and  $\mathbf{Q}_g[\phi_g] < \epsilon$  then                ▷ Inconsistent State #2
23:      $\phi_l \leftarrow 1.0$ 
24:      $\rho_l \leftarrow \mathbf{Q}_l[\phi_l \rho_l] / 1.0$ 
25:      $\phi_g \leftarrow 0.0$ 
26:      $\rho_g \leftarrow \text{NULL}$ 
27:   else
28:      $\phi_l \leftarrow \mathbf{Q}_l[\phi_l]$ 
29:      $\rho_l \leftarrow \mathbf{Q}_l[\phi_l \rho_l] / \mathbf{Q}_l[\phi_l]$                         ▷ Nominal case with phase distribution defined by  $\mathbf{Q}_l$ 
30:      $\phi_g \leftarrow \mathbf{Q}_l[\phi_g]$ 
31:      $\rho_g \leftarrow \mathbf{Q}_g[\phi_g \rho_g] / \mathbf{Q}_l[\phi_g]$ 
32:   end if
33:    $\mathbf{Q}^{\text{new}} \leftarrow \{\phi_l, \phi_g, \phi_l \rho_l, \phi_g \rho_g, \phi_l \rho_l \mathbf{u}_l, \phi_g \rho_g \mathbf{u}_g, \phi_g \rho_g Y_g, \phi_l \rho_l E_l, \phi_g \rho_g E_g\}$ 
34:   return  $\mathbf{Q}^{\text{new}}$ 
35: end function

```

- On line 2, all quantities are advected as a function of the gas phase advective velocity $u_{a,g}$ from time level a to time level $a_{1/6}$ using the CISL algorithm, resulting in an intermediate state vector $\mathbf{Q}_g^{a_{1/6}}$. The intermediate state vector $\mathbf{Q}_g^{a_{1/6}}$ contains the volume fraction, mass, momentum, and total energy for both phases (and both species in the gas phase), advected via the *gas-phase* advective velocity $u_{a,g}$.
- Next, on line 3, all quantities are advected as a function of the liquid-phase advective velocity $u_{a,l}$ from time level a to time level $a_{1/6}$ using the CISL algorithm, resulting in a second intermediate state vector $\mathbf{Q}_l^{a_{1/6}}$. The intermediate state vector $\mathbf{Q}_l^{a_{1/6}}$ contains the volume fraction, mass, momentum, and total energy for both phases (and both species in the gas phase), advected via the *liquid phase* advective velocity $u_{a,l}$.
- On line 4, intermediate state vectors $\mathbf{Q}_g^{a_{1/6}}$ and $\mathbf{Q}_l^{a_{1/6}}$ are conservatively combined into a single state vector $\mathbf{Q}^{a_{1/6}}$ as follows:
 - Liquid phase conserved quantities (mass $\mathbf{Q}_l[\phi_l \rho_l]$, energy $\mathbf{Q}_l[\phi_l \rho_l E_l]$, and momentum $\mathbf{Q}_l[\phi_l \rho_l \mathbf{u}_l]$) are populated by the corresponding quantities contained in intermediate state vector $\mathbf{Q}_l^{a_{1/6}}$. The bracket notation denotes the quantity enclosed by the brackets to be contained within the preceding vector. For example, $\mathbf{Q}_l[\phi_l \rho_l \mathbf{u}_l]$ on line 11 denotes the liquid phase momentum $\phi_l \rho_l \mathbf{u}_l$ to be contained in vector $\mathbf{Q}_l$, where $\mathbf{Q}_l$ results from performing advection with the liquid phase advective velocity. On lines 11-12, the liquid velocity and total energy density are computed from the conserved quantities.
 - Gas phase conserved quantities (mass $\mathbf{Q}_g[\phi_g \rho_g]$, energy $\mathbf{Q}_g[\phi_g \rho_g E_g]$, and momentum $\mathbf{Q}_g[\phi_g \rho_g \mathbf{u}_g]$) are populated by the corresponding quantities contained in intermediate state vector $\mathbf{Q}_g^{a_{1/6}}$. Bracket notation applies to gas phase quantities in the same way as liquid phase quantities. On lines 13-15, the gas phase velocity, total energy density, and vapor mass fraction are computed from the conserved quantities.
- Completing step 3 defines the transport of gas phase conserved quantities by the gas phase advective velocity, and liquid phase conserved quantities by the liquid phase advective velocity. Having advected all conserved quantities, the task of step 4 is to determine a spatial distribution of phase that is consistent with the spatial distribution of mass defined in step 3. Consistency is achieved when two constraints are satisfied by the chosen VOF field: (1) every cell i defined to contain a non-zero mass of phase m in step 3 is assigned a non-zero VOF, $\phi_m > 0$, in step 4; and (2) every cell i defined to contain zero mass of phase m in step 3 is assigned a VOF of zero in step 4, $\phi_m = 0$. Nominally,

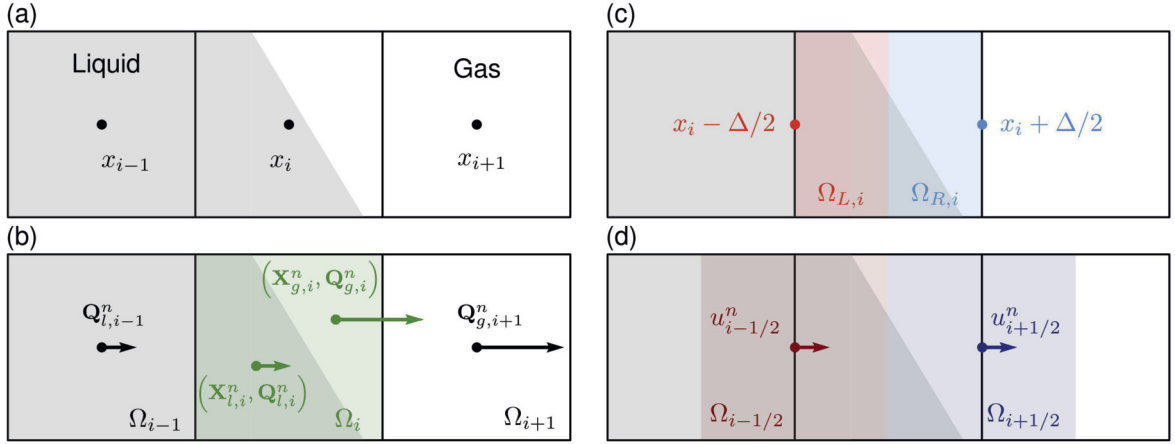


Fig. 3. Schematic illustrating the geometric quantities of cell i and its neighbors $i-1$ and $i+1$ relevant to updating the face-centered velocities via advection: (a) mixed liquid/gas cell i and neighboring single-phase cells; (b) region Ω_i corresponding to cell i highlighted in green, and quantities of interest $\mathbf{Q}$ indicated at phase-specific centers of mass; (c) left-side region $\Omega_{L,i}$ (light red) and right-side region $\Omega_{R,i}$ (light blue) of cell i ; (d) face-centered regions $\Omega_{i-1/2}$ (dark red) and $\Omega_{i+1/2}$ (dark blue), along with the associated face-centered, single-fluid velocities.

the volumetric distribution of phase can be defined by the VOFs ϕ_m contained in intermediate vector $\mathbf{Q}_i^{a1/6}$ (the intermediate vector produced as a function of the liquid phase advective velocity). This nominal operation, along with the computation of density from mass and VOF, appears on lines 27-31. However, the phase-specific advective velocities do not form a tessellation when combined, and this nominal operation does not always produce a consistent state. Possible inconsistencies are defined and accounted for on lines 17-26:

- (a) Lines 17-21 identify Inconsistent State #1, where a cell contains liquid mass, $\mathcal{Q}_l[\phi_l \rho_l] > 0$, and gas mass, $\mathcal{Q}_g[\phi_g \rho_g] > 0$. However, the gas phase VOF provided by the nominal operation is zero, $\mathcal{Q}_l[\phi_g] = 0$. In this event, the liquid and gas volumes for the cell are defined by $\mathcal{Q}_g[\phi_g]$ and $\mathcal{Q}_g[\phi_l]$, rather than $\mathcal{Q}_l[\phi_g]$ and $\mathcal{Q}_l[\phi_l]$. Rather than comparison against zero, the practical implementation considers a small number, $\epsilon = 10^{-10}$, to account for numerical round-off error and very small volume fractions.
 - (b) Lines 22-26 identify Inconsistent State #2, where a cell contains liquid mass, $\mathcal{Q}_l[\phi_l \rho_l] > 0$, a nominal liquid phase VOF less than one, $\mathcal{Q}_l[\phi_l] < 1$, and zero gas phase mass $\mathcal{Q}_g[\phi_g \rho_g] = 0$. In this event, the liquid phase is expanded to fill the entire computational cell.
5. The combined state vector $\mathbf{Q}^{a1/6}$ is finally constructed and returned on lines 33-34.
 6. Steps 1-5 are repeated for the remaining CISL sub-steps according to Algorithm 1.

4.3.2. Cell-centered and face-centered quantities

A computational cell i containing gray liquid and white gas is shown in Fig. 3 panel (a), along with neighboring cells $j = i-1$ and $j = i+1$. Cell i corresponds to region Ω_i , defined by

$$\Omega_i = (x, y, z) \in [x_i - \Delta/2, x_i + \Delta/2] \times [y_i - \Delta/2, y_i + \Delta/2] \times [z_i - \Delta/2, z_i + \Delta/2], \quad (31)$$

where x_i , y_i , and z_i are the coordinates of the cell center, and Δ is the uniform Cartesian mesh spacing. Region Ω_i appears highlighted in green in Fig. 3 panel (b), and contains liquid with center of mass $\mathbf{x}_{l,i}^n$ and gas with center of mass $\mathbf{x}_{g,i}^n$. The liquid and gas quantities of interest, $\mathbf{Q}_{l,i}^n$ and $\mathbf{Q}_{g,i}^n$, are stored at their respective centers of mass. Phase-specific velocity vectors, $\mathbf{u}_{l,i}^n$ and $\mathbf{u}_{g,i}^n$, are also located at the centers of mass (transported in $\mathbf{Q}_{l,i}^n$ and $\mathbf{Q}_{g,i}^n$ via the momentum and mass).

The mass M_i^n and momentum $\mathbf{P}_i^n$ contained within cell i at time level n are given by

$$M_i^n = \sum_m \rho_m^n \phi_m^n V, \quad (32)$$

and

$$\mathbf{P}_i^n = \sum_m \mathbf{u}_m^n \rho_m^n \phi_m^n V, \quad (33)$$

where $V = \Delta^3$ is the volume of region Ω_i . From Eq. (29), the corresponding single-fluid velocity is

$$\mathbf{u}_i^n = \frac{\mathbf{P}_i^n}{M_i^n}. \quad (34)$$

The control volumes associated with neighboring cells $i - 1$ and $i + 1$ are defined by regions Ω_{i-1} and Ω_{i+1} . These cells contain only a single phase, and the center of mass of the relevant phase is coincident with the cell-center.

Fig. 3 panel (c) decomposes cell i into left-half region $\Omega_{L,i}$ (light red), and right-half region $\Omega_{R,i}$ (light blue). The masses contained in the left-half and right-half regions at the pre-advection time level n are given by

$$M_{L,i}^n = \sum_m \int_{\Omega_{L,i}} \rho_m^n \chi_m^n d\Omega, \quad (35)$$

and

$$M_{R,i}^n = \sum_m \int_{\Omega_{R,i}} \rho_m^n \chi_m^n d\Omega, \quad (36)$$

respectively, where χ_m^n is the discrete phase indicator corresponding to the PLIC reconstruction within the specified region at time level n . For region $\Omega_{L,i}$ at time level n , the phase indicator χ_m^n is computed from

$$\chi_m^n = \begin{cases} 1 & \mathbf{n}_{m,i} \cdot (\mathbf{x} - \mathbf{x}_i) + b_{m,i}^n \geq 0 \text{ and } \mathbf{x} \in \Omega_{L,i}. \\ 0 & \text{else.} \end{cases} \quad (37)$$

An analogous expression defines $\Omega_{R,i}$.

Fig. 3 panel (d) shows the face-centered regions $\Omega_{i-1/2}$ (dark red) and $\Omega_{i+1/2}$ (dark blue) and the corresponding face-centered, single-fluid velocities $u_{i-1/2}$ and $u_{i+1/2}$. The face-centered velocities are spatially staggered, with a single velocity component defined normal to the cell face. The pre-advection, face-centered momentums $P_{i-1/2}^n$ and $P_{i+1/2}^n$ are computed from

$$P_{i-1/2}^n = \sum_m \int_{\Omega_{i-1/2}} u_{i-1/2}^n \rho_m^n \chi_m^n d\Omega \quad (38)$$

and

$$P_{i+1/2}^n = \sum_m \int_{\Omega_{i+1/2}} u_{i+1/2}^n \rho_m^n \chi_m^n d\Omega. \quad (39)$$

The face-centered masses $M_{i-1/2}^n$ and $M_{i+1/2}^n$ are defined analogously to Eq. (35) and Eq. (36) by integrating over the relevant regions $\Omega_{i-1/2}$ and $\Omega_{i+1/2}$.

Fig. 4 shows the interfacial system from Fig. 3 in the pre- and post-advection states (in the left and right columns, respectively). The top row shows the interface position and the state of the cell-centered, single-fluid momentum. Advancing to time level a via the CISL advection procedure, the single-fluid momentum in cell i changes by a quantity ΔP_i . Cell i momentum change corresponds to changes in mass and single-fluid velocity (shown in the second row, with each half-cell mass shown in a different color) according to

$$\Delta P_i = u_i^a (M_{L,i}^a + M_{R,i}^a) - u_i^n (M_{L,i}^n + M_{R,i}^n). \quad (40)$$

The bottom row shows the state of the face-centered, single-fluid velocities, as well as the face-centered masses. Unlike all other quantities shown in Fig. 4, the post-advection, face-centered, single-fluid velocities are not provided by the CISL advection procedure and require a closure strategy. These unclosed terms have been circled by red dashed lines.

4.3.3. Face-centered velocity update

After completing the advection procedure defined in Section 4.3.1 and obtaining $\mathbf{Q}_i^a$, the face-centered velocities must also be updated to the post-advection time level. In order to ensure conservation, we seek a relationship between the cell-centered momentum change ΔP_i and the face-centered momentum change $\Delta P_{i-1/2}$, where

$$\begin{aligned} \Delta P_{i-1/2} &= P_{i-1/2}^a - P_{i-1/2}^n \\ &= u_{i-1/2}^a M_{i-1/2}^a - u_{i-1/2}^n M_{i-1/2}^n. \end{aligned} \quad (41)$$

Because the CISL advection scheme is explicit, we assume changes in face-centered momentums are related to changes in adjacent cell-centered momentums according to

$$\Delta P_{i-1/2} = \alpha_{R,i-1} \Delta P_{i-1} + \alpha_{L,i} \Delta P_i, \quad (42)$$

where α is a distribution coefficient. Momentum conservation is achieved when the distribution coefficients satisfy

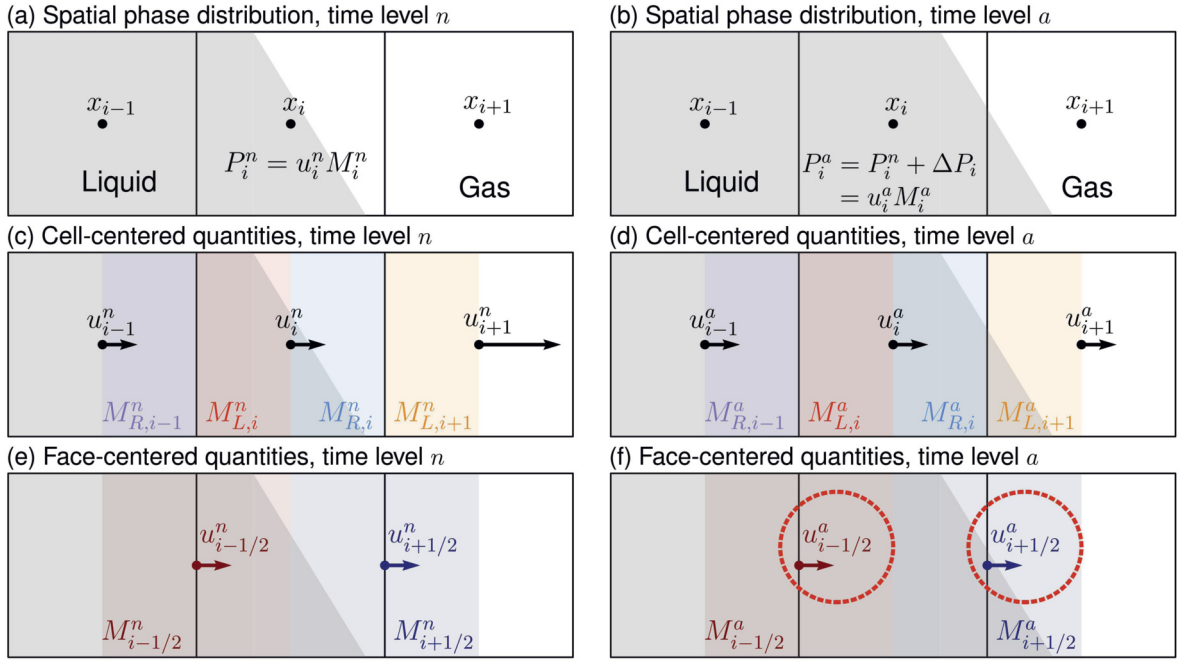


Fig. 4. Evolution of a liquid/gas interface translating from left-to-right from time level n to the post-advection time level a : (a-b) change in phase distribution and cell-centered momentum as a function of CISL advection; (c-d) cell-centered single-fluid velocities and left-half and right-half masses at time levels n and a ; (e-f) face-centered masses and face-centered, single-fluid velocities at time levels n and a . The face-centered, single-fluid velocities at time level a (circled by red dashes) are not provided by the CISL procedure, and require closure.

$$\alpha_{L,i} + \alpha_{R,i} = 1 \quad (43)$$

for each cell i .

In order to determine an appropriate expression for the distribution coefficients, we consider the second-order Taylor series expansion of momentum in time, given by

$$P^a = P^n + \frac{dP^n}{dt} \Delta t + \frac{1}{2} \frac{d^2 P^n}{dt^2} \Delta t^2 + \mathcal{O}(\Delta t^3). \quad (44)$$

Expanding the first three terms on the right-hand-side of Eq. (44) into the product of mass and velocity, followed by application of the chain rule, returns

$$P^n + \frac{dP^n}{dt} \Delta t + \frac{1}{2} \frac{d^2 P^n}{dt^2} \Delta t^2 = u^n M^n + u^n \frac{dM^n}{dt} \Delta t + M^n \frac{du^n}{dt} \Delta t + \left(2 \frac{dM^n}{dt} \frac{du^n}{dt} + u^n \frac{d^2 M^n}{dt^2} + M^n \frac{d^2 u^n}{dt^2} \right) \frac{\Delta t^2}{2}. \quad (45)$$

As illustrated in Fig. 4, cell-centered velocities and masses are known at time levels n and a . With two available time levels, we neglect the final two second-order temporal terms on the right-hand-side of Eq. (45). Substitution of the remaining terms on the right-hand-side of Eq. (45) into Eq. (44), followed by application of the result to face $i - 1/2$, results in an approximation of the time level a face-centered momentum, given by

$$P_{i-1/2}^a = u_{i-1/2}^n M_{i-1/2}^n + u_{i-1/2}^n \frac{dM_{i-1/2}^n}{dt} \Delta t + M_{i-1/2}^n \frac{du_{i-1/2}^n}{dt} \Delta t + \frac{dM_{i-1/2}^n}{dt} \frac{du_{i-1/2}^n}{dt} \Delta t^2 + \mathcal{O}(\Delta t^2). \quad (46)$$

Equation (46) includes four terms on the right-hand-side, which are identified by the following naming convention (from left-to-right): the pre-advection, face-centered momentum, the mass change contribution, the velocity change contribution, and the mixed mass/velocity contribution. The closure strategy chosen for these terms involves discretization as a function of known cell-centered velocities and half-cell masses at time levels n and a , such that the resultant form satisfies Eq. (42) and Eq. (43).

Using the known half-cell masses at time levels n and a (illustrated in Fig. 4), the temporal derivative of face-centered mass is approximated with first-order accuracy according to

$$\begin{aligned}\frac{dM_{i-1/2}^n}{dt} &= \frac{M_{i-1/2}^a - M_{i-1/2}^n}{\Delta t} + \mathcal{O}(\Delta t) \\ &= \frac{(M_{L,i}^a - M_{L,i}^n) + (M_{R,i-1}^a - M_{R,i-1}^n)}{\Delta t} + \mathcal{O}(\Delta t).\end{aligned}\quad (47)$$

The temporal derivative of face-centered velocity requires interpolation of the cell-centered velocities. In this work, an equal weighting is applied to the i and $i - 1$ velocity changes, corresponding to

$$\frac{du_{i-1/2}^n}{dt} \approx \frac{(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)}{2\Delta t}.\quad (48)$$

Using Eq. (47) and Eq. (48), the mass change contribution and the mixed mass/velocity contribution are discretely approximated by

$$u_{i-1/2}^n \frac{dM_{i-1/2}^n}{dt} \Delta t \approx u_{i-1/2}^n [(M_{L,i}^a - M_{L,i}^n) + (M_{R,i-1}^a - M_{R,i-1}^n)],\quad (49)$$

and

$$\frac{dM_{i-1/2}^n}{dt} \frac{du_{i-1/2}^n}{dt} \Delta t^2 \approx \frac{1}{2} [(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)] [(M_{L,i}^a - M_{L,i}^n) + (M_{R,i-1}^a - M_{R,i-1}^n)],\quad (50)$$

respectively.

Substituting the discretized forms of the mass change contribution, Eq. (49), and the mixed mass/velocity contribution, Eq. (50), into Eq. (46) returns

$$\begin{aligned}P_{i-1/2}^a &= u_{i-1/2}^n M_{i-1/2}^n + \\ &\quad u_{i-1/2}^n [(M_{L,i}^a - M_{L,i}^n) + (M_{R,i-1}^a - M_{R,i-1}^n)] + \\ &\quad \frac{1}{2} [(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)] [(M_{L,i}^a - M_{L,i}^n) + (M_{R,i-1}^a - M_{R,i-1}^n)] + \\ &\quad M_{i-1/2}^n \frac{du_{i-1/2}^n}{dt} \Delta t.\end{aligned}\quad (51)$$

After consolidating mass changes associated with i and $i - 1$, we arrive at the following expression:

$$\begin{aligned}P_{i-1/2}^a &= u_{i-1/2}^n M_{i-1/2}^n + \\ &\quad \left(u_{i-1/2}^n + \frac{1}{2} [(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)] \right) (M_{L,i}^a - M_{L,i}^n) + \\ &\quad \left(u_{i-1/2}^n + \frac{1}{2} [(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)] \right) (M_{R,i-1}^a - M_{R,i-1}^n) + \\ &\quad M_{i-1/2}^n \frac{du_{i-1/2}^n}{dt} \Delta t.\end{aligned}\quad (52)$$

The second and third factors on the right-hand-side are identified as the preliminary momentum contributions, defined by

$$\delta P_{L,i} = \left(u_{i-1/2}^n + \frac{1}{2} [(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)] \right) (M_{L,i}^a - M_{L,i}^n),\quad (53)$$

and

$$\delta P_{R,i-1} = \left(u_{i-1/2}^n + \frac{1}{2} [(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)] \right) (M_{R,i-1}^a - M_{R,i-1}^n).\quad (54)$$

Equations (53) and (54) share a common factor that approximates the time level a face-centered velocity,

$$\begin{aligned}u_{i-1/2}^a &= u_{i-1/2}^n + \frac{du_{i-1/2}^n}{dt} \Delta t + \mathcal{O}(\Delta t) \\ &\approx u_{i-1/2}^n + \frac{1}{2} [(u_i^a - u_i^n) + (u_{i-1}^a - u_{i-1}^n)],\end{aligned}\quad (55)$$

where the discrete temporal derivative of face-centered velocity is defined by Eq. (48). The preliminary momentum contributions therefore approximate the time level a face-centered velocity multiplied by the change in face-centered mass,

$$\delta P_{L,i} + \delta P_{R,i-1} \approx u_{i-1/2}^a \frac{dM_{i-1/2}^n}{dt} \Delta t, \quad (56)$$

where $\delta P_{L,i}$ provides the mass change contribution from cell i , and $\delta P_{R,i-1}$ from cell $i-1$. Imposing a similar decomposition into cell i and $i-1$ contributions on the unclosed velocity change contribution,

$$M_{i-1/2}^n \frac{du_{i-1/2}^n}{dt} \Delta t \approx M_{R,i-1}^n \frac{du_{i-1}^n}{dt} \Delta t + M_{L,i}^n \frac{du_i^n}{dt} \Delta t, \quad (57)$$

leads to an alternative form of Eq. (52),

$$P_{i-1/2}^a = u_{i-1/2}^n M_{i-1/2}^n + \delta P_{L,i} + \delta P_{R,i-1} + M_{R,i-1}^n \frac{du_{i-1}^n}{dt} \Delta t + M_{L,i}^n \frac{du_i^n}{dt} \Delta t, \quad (58)$$

where the velocity change contributions from cells i and $i-1$ remain unclosed.

Manipulating Eq. (58) into a function of the face-centered momentum change, substituting the right-hand-side of Eq. (41), moving the $i-1$ -dependent terms to the left-hand-side, and moving the i -dependent terms to the right-hand side, returns

$$M_{R,i-1}^n \frac{du_{i-1}^n}{dt} \Delta t + (\alpha_{R,i-1} \Delta P_{i-1} - \delta P_{R,i-1}) = M_{L,i}^n \frac{du_i^n}{dt} \Delta t + (\alpha_{L,i} \Delta P_i - \delta P_{L,i}). \quad (59)$$

Equation (59) can only be satisfied if both sides equal zero (the mass terms on the left-hand-side are independent of the mass terms on the right-hand-side). Acknowledging this condition, we permute the indices by one,

$$M_{R,i}^n \frac{du_i^n}{dt} \Delta t + (\alpha_{R,i} \Delta P_i - \delta P_{R,i}) = M_{L,i+1}^n \frac{du_{i+1}^n}{dt} \Delta t + (\alpha_{L,i+1} \Delta P_{i+1} - \delta P_{L,i+1}), \quad (60)$$

equate the left-hand-side of Eq. (60) and the right-hand-side of Eq. (59), and rearrange into the following form:

$$M_{L,i}^n \frac{du_i^n}{dt} \Delta t + M_{R,i}^n \frac{du_i^n}{dt} \Delta t = (\alpha_{L,i} \Delta P_i - \delta P_{L,i}) + (\alpha_{R,i} \Delta P_i - \delta P_{R,i}). \quad (61)$$

Applying the conservation constraint of Eq. (43) to Eq. (61) returns

$$\begin{aligned} M_{L,i}^n \frac{du_i^n}{dt} \Delta t + M_{R,i}^n \frac{du_i^n}{dt} \Delta t &= \Delta P_i - \delta P_{L,i} - \delta P_{R,i} \\ &= R_i \end{aligned} \quad (62)$$

where R_i is the residual momentum change. The velocity change contributions are closed with the following mass-fraction-based weighting:

$$\begin{aligned} M_{L,i}^n \frac{du_i^n}{dt} \Delta t &= R_i \frac{M_{L,i}^a}{M_i^a} \\ &= R_{L,i} \end{aligned} \quad (63)$$

and

$$\begin{aligned} M_{R,i}^n \frac{du_i^n}{dt} \Delta t &= R_i \frac{M_{R,i}^a}{M_i^a} \\ &= R_{R,i}. \end{aligned} \quad (64)$$

Defining the velocity change contributions finalizes closure of the time level a face-centered momentum,

$$P_{i-1/2}^a = M_{i-1/2}^n u_{i-1/2}^n + (\delta P_{L,i} + \delta P_{R,i-1}) + (R_{L,i} + R_{R,i-1}), \quad (65)$$

and the face-centered, single-fluid velocity,

$$u_{i-1/2}^a = \frac{M_{i-1/2}^n u_{i-1/2}^n + (\delta P_{L,i} + \delta P_{R,i-1}) + (R_{L,i} + R_{R,i-1})}{M_{i-1/2}^a}. \quad (66)$$

Updating the face-centered velocities according to Eq. (66) completes the advancement to post-advection time level a .

4.4. Diffusion and phase change

Imposing the effects of diffusion and phase change advances the post-advection state $\mathbf{Q}^a$ to the post-diffusion state $\mathbf{Q}^d$ via Eq. (24). This section defines the mixed time level diffusion and phase change vectors $\mathbf{D}_{exp}^a$ and $\mathbf{D}_{imp}^d$.

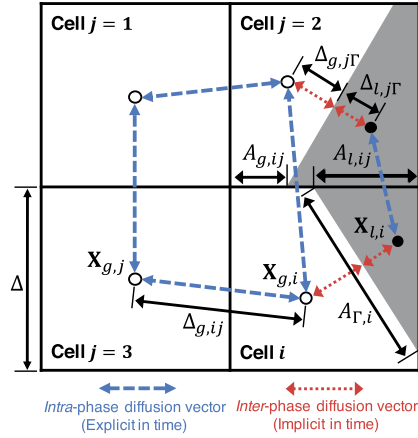


Fig. 5. Geometric reconstruction near a liquid/gas interface, illustrating the geometric quantities required to compute diffusion and phase change. Blue dashed lines indicate diffusion paths discretized explicitly, and red dotted lines indicate diffusion paths discretized implicitly.

4.4.1. Intra-phase diffusion

Intra-phase diffusion, defined as diffusion within a single phase, occurs between cell i and its N_j neighboring cells $j = 1, 2, \dots, N_j - 1, N_j$, where $N_j = 6$ in three-dimensions. Diffusion of heat and species mass fraction is performed in the sharp interface context by leveraging the geometric quantities introduced in Section 4.2. Viscous diffusion and viscous heating are imposed in a lumped finite-volume formulation identical to Jemison et al. [36]: the viscous stress tensor is evaluated at cell faces as a function of time level a cell-centered, single-fluid velocity, and the face-centered viscosity; the face-centered viscosity is evaluated as the volume-weighted average of the phases present in the face-centered control volume; divergences of the viscous stress tensor are conservatively applied via a second-order finite-volume discretization.

The rate of thermal diffusion between the gas in cell i and the gas in neighboring cell j is quantified by

$$D_{g,ij}^a = k_g \frac{(T_{g,j}^a - T_{g,i}^a)}{\|\mathbf{X}_{g,j} - \mathbf{X}_{g,i}\|} A_{g,ij} + \rho_{g,ij}^a \mathcal{D}_{ij}^a (h_{v,ij}^a - h_{nc,ij}^a) \frac{(Y_j^a - Y_i^a)}{\|\mathbf{X}_{g,j} - \mathbf{X}_{g,i}\|} A_{g,ij}, \quad (67)$$

where the first and second terms on the right-hand-side account for thermal conduction and enthalpy transport due to mass diffusion, respectively. The density, diffusivity, vapor enthalpy, and non-condensable gas enthalpy include subscripts ij to indicate that these quantities are defined by the average value of cells i and j . For example, the density $\rho_{g,ij}^a$ is defined by

$$\rho_{g,ij}^a = \frac{\rho_{g,i}^a + \rho_{g,j}^a}{2}. \quad (68)$$

The diffusion length scale for both terms in Eq. (67) is defined as the distance between the gas phase centers of mass in cells i and j , $\|\mathbf{X}_{g,j} - \mathbf{X}_{g,i}\|$, and the diffusion flux operates on the gas phase wetted area between cells i and j , $A_{g,ij}$. The gas phase diffusion paths between centers of mass $\mathbf{X}_{g,i}$ and $\mathbf{X}_{g,j}$ are illustrated in Fig. 5, appearing as blue dashed lines.

Cell i exchanges energy according to Eq. (67) with each of its N_j neighbors. Summing the contributions from each neighbor and dividing the result by the cell volume V returns the intra-phase thermal diffusion forcing term for the gas phase,

$$D_{g,i}^a = \frac{1}{V} \sum_{j=1}^{N_j} \mathbf{D}_j \left[k_g \frac{(T_{g,j}^a - T_{g,i}^a)}{\|\mathbf{X}_{g,j} - \mathbf{X}_{g,i}\|} A_{g,ij} + \rho_{g,ij}^a \mathcal{D}_{ij}^a (h_{v,ij}^a - h_{nc,ij}^a) \frac{(Y_j^a - Y_i^a)}{\|\mathbf{X}_{g,j} - \mathbf{X}_{g,i}\|} A_{g,ij} \right]. \quad (69)$$

The column vector $\mathbf{D}_j$ contains entries of 1 or -1 depending on the location of j relative to i , such that Eq. (69) forms a divergence operator. The expressions for species transport in the gas phase and thermal diffusion in the liquid phase are defined analogously:

$$D_{v,i}^a = \frac{1}{V} \sum_{j=1}^{N_j} \mathbf{D}_j \rho_{g,ij}^a \mathcal{D}_{ij}^a \frac{(Y_j^a - Y_i^a)}{\|\mathbf{X}_{g,j} - \mathbf{X}_{g,i}\|} A_{g,ij}, \quad (70)$$

and

$$D_{l,i}^a = \frac{1}{V} \sum_{j=1}^{N_j} \mathbf{D}_j k_l \frac{(T_{l,j}^a - T_{l,i}^a)}{\|\mathbf{X}_{l,j} - \mathbf{X}_{l,i}\|} A_{l,ij}. \quad (71)$$

The cell-centered, single-fluid velocity is advanced to the post-diffusion time d for all cells according to

$$\mathbf{u}_i^d = \mathbf{u}_i^a + \frac{(\nabla \cdot \boldsymbol{\tau}^a)_i}{\rho_i^a} \Delta t, \quad (72)$$

expressed here in semi-discrete form [36]. Advancing the energy and vapor mass fraction requires consideration as to whether the cell contains a single phase or multiple phases. The energy of single phase cells is advanced to time level d as a function of viscous effects and thermal diffusion according to

$$E_{m,i}^d = E_{m,i}^a + \frac{(\nabla \cdot (\mathbf{u}^a \cdot \boldsymbol{\tau}^a))_i}{\rho_i^a} \Delta t + \frac{D_{m,i}^a}{\rho_i^a} \Delta t. \quad (73)$$

Mixed-phase cells are advanced only as a function of viscous effects, because inter-phase diffusion has not yet been defined, resulting in an intermediate energy $E_{m,i}^{d*}$,

$$E_{m,i}^{d*} = E_{m,i}^a + \frac{(\nabla \cdot (\mathbf{u}^a \cdot \boldsymbol{\tau}^a))_i}{\rho_i^a} \Delta t. \quad (74)$$

Lastly, the vapor mass fraction of entirely gas-filled cells is advanced according to

$$Y_i^d = Y_i^a + \frac{D_{v,i}^a}{\rho_i^a} \Delta t. \quad (75)$$

Each of the intra-phase diffusion rates defined in this section are explicit functions of the post-advection state $\mathbf{Q}^a$, and quantify the rates of diffusion within like-phases in neighboring cells: inter-cell thermal diffusion in the gas phase $D_{g,i}^a$, defined by Eq. (69), inter-cell diffusion of species mass fraction in the gas phase $D_{v,i}^a$, defined by Eq. (70), and inter-cell conduction in the liquid phase $D_{l,i}^a$, defined by Eq. (71). Alternatively, combining the present notation with that of Fig. 1 produces the analogous mathematical statement:

$$\mathbf{D}_{exp}^a = f(D_{l,i}^a, D_{g,i}^a, D_{v,i}^a). \quad (76)$$

Diffusion and phase change also occur between the liquid and gas via transport across the interface. Quantifying inter-phase diffusion and phase change is required to complete the time advancement to post-diffusion time level d .

4.4.2. Inter-phase diffusion and phase change

Inter-phase diffusion occurs between the liquid and gas within cell i via the interface. Interfacial diffusion is coupled to the rate of phase change via Eq. (14) and Eq. (17). The area specific phase change rate $\dot{m}$, the interface vapor mass fraction Y_Γ , the interface temperature T_Γ , and the liquid and gas temperatures are coupled by a first-order discretization inspired by the “MB” method of Palmore and Desjardins [13], which is based on the fixed-point formulation of Ma and Bothe [11]. In this framework, discretization of Eq. (14) is given by

$$-\hat{H}^d \dot{m}^d = \frac{k_g (T_\Gamma^d - T_g^d)}{\|\mathbf{X}_\Gamma^a - \mathbf{X}_g^a\|} + \frac{k_l (T_\Gamma^d - T_l^d)}{\|\mathbf{X}_\Gamma^a - \mathbf{X}_l^a\|}. \quad (77)$$

Because all quantities in this section correspond to a single cell i , the subscript i has been omitted to simplify the notation. The centers of mass for the gas and liquid phases within the computational cell are therefore denoted simply by $\mathbf{X}_g$ and $\mathbf{X}_l$, while $\mathbf{X}_\Gamma$ denotes the flux evaluation point on the interface. The diffusion paths connecting the liquid and gas centers of mass with the interface appear as red dotted lines in Fig. 5. The thermodynamic properties are defined implicitly at time level d , while the geometric properties are defined explicitly at the advection time level a . An identical discretization strategy couples the evaporation rate to the interfacial mass diffusion rate via Eq. (17):

$$\dot{m}^d = \frac{\rho_g^d \mathcal{D}}{1 - Y_\Gamma^d} \left[\frac{(Y_\Gamma^d - Y^d)}{\|\mathbf{X}_\Gamma^a - \mathbf{X}_g^a\|} \right]. \quad (78)$$

The interfacial vapor mass fraction Y_Γ^d is assumed to be the equilibrium state defined by the interface pressure and temperature. The interfacial vapor mass fraction is defined by a material-specific function $\mathcal{Y}$ of the interface temperature and the gas density:

$$Y_{\Gamma}^d = \mathcal{Y}(T_{\Gamma}^d, \rho_g^d). \quad (79)$$

The mass transferred across the liquid/gas interface within the computational cell over time Δt is given by

$$\Delta m^d = \dot{m}^d A_{\Gamma}^a \Delta t, \quad (80)$$

where A_{Γ}^a is the interfacial surface area in the computational cell, computed from the PLIC reconstruction and identified in Fig. 5. For notational convenience, we introduce the interfacial area density,

$$\bar{A}_{\Gamma}^a = \frac{A_{\Gamma}^a}{V}, \quad (81)$$

where V is the volume of the computational cell. Over the course of the time step Δt , mass transfer across the interface modifies the liquid mass contained in the computational cell according to

$$\phi_l^d \rho_l^d = \phi_l^a \rho_l^a - \dot{m}^d \bar{A}_{\Gamma}^a \Delta t. \quad (82)$$

The gas phase mass at time level d is analogously

$$\phi_g^d \rho_g^d = \phi_g^a \rho_g^a + \dot{m}^d \bar{A}_{\Gamma}^a \Delta t. \quad (83)$$

According to Eqs. (82) and (83), changes in either density or volume fraction produce changes in mass as time advances from time level a to time level d . In the liquid phase, the change in mass is imposed by assuming the phase change dynamic does not modify the liquid density (enforcing $\rho_l^d = \rho_l^a$), thus imparting all mass change via the volume fraction:

$$\phi_l^d = \phi_l^a - \frac{\dot{m}^d \bar{A}_{\Gamma}^a \Delta t}{\rho_l^d}. \quad (84)$$

The gaseous VOF at time level d must therefore satisfy

$$\phi_g^d = \phi_g^a + \frac{\dot{m}^d \bar{A}_{\Gamma}^a \Delta t}{\rho_l^d}, \quad (85)$$

imposed by the volume conservation identity, $\phi_l + \phi_g = 1$. The gas phase undergoes an additional change in density due to evaporation or condensation of vapor, constrained by conservation of mass at time level d , expressed by Eq. (83). The addition or removal of vapor changes the energy density of the gas phase contained in the cell, because the properties of the evaporating vapor and the cell-averaged gas inevitably differ.

The energy density associated with the fluid transferring from one phase to the other is denoted by $\mathcal{E}$. In the case of evaporation, mass transfers from the liquid phase to the gas phase at an area-specific rate of $\dot{m}$. The energy contained in the fluid element in the liquid is denoted by $\mathcal{E}_l$, and the energy of the element after it has entered the vapor phase is denoted by $\mathcal{E}_v$. The liquid phase energy $\mathcal{E}_l$ is the total energy of the liquid phase evaluated at the interface state

$$\mathcal{E}_l^d = e_l^d(T_{\Gamma}^d, \rho_l^a) + \frac{1}{2} \mathbf{u}^d \cdot \mathbf{u}^d. \quad (86)$$

From the definition of the enthalpy of evaporation, the total energy of the vapor phase undergoing phase transfer is given by

$$\mathcal{E}_v^d = \mathcal{E}_l^d + \hat{H}^d. \quad (87)$$

Incorporating the explicit intra-phase diffusion, D_g^a defined by Eq. (69) in Section 4.4.1, returns an implicit expression for the gas phase total energy at time level d ,

$$\rho_g^d E_g^d \phi_g^d = \rho_g^a E_g^{d*} \phi_g^a + \mathcal{E}_v^d \dot{m}^d \bar{A}_{\Gamma}^a \Delta t + k_g (T_{\Gamma}^d - T_g^d) \bar{A}_{\Gamma}^a \Delta_{g,\Gamma}^{-1} \Delta t + D_g^a \Delta t, \quad (88)$$

where the inverse of gas phase interfacial distance has been expressed by

$$\Delta_{g,\Gamma}^{-1} = \frac{1}{\|\mathbf{x}_{\Gamma}^a - \mathbf{x}_g^a\|}. \quad (89)$$

The right-hand-side of Eq. (88) includes four terms: the first is the total energy at time level a ; the second is the total energy of the newly evaporated vapor that has been added to the gas phase mixture within the cell; the third term is the heat conduction through the interface; and the fourth is the explicit energy exchange via heat and enthalpy diffusion with the gas phase in neighboring cells. Similarly, heat conduction through the interface, liquid mass change, and exchange of energy with liquid in neighboring cells define the liquid total energy at time level d according to

$$\rho_l^d E_l^d \phi_l^d = \rho_l^a E_l^{d*} \phi_l^a - \mathcal{E}_l^d \dot{m}^d \bar{A}_\Gamma^a \Delta t + k_l (T_\Gamma^d - T_l^d) \bar{A}_\Gamma^a \Delta_{l,\Gamma}^{-1} \Delta t + D_l^a \Delta t, \quad (90)$$

where

$$\Delta_{l,\Gamma}^{-1} = \frac{1}{\|\mathbf{X}_\Gamma^a - \mathbf{X}_l^a\|}. \quad (91)$$

Lastly, the addition or removal of vapor due to phase change, and the exchange of vapor with neighboring cells via diffusion, defines the vapor mass fraction at time level d :

$$\rho_g^d Y_g^d \phi_g^d = \rho_g^a Y_g^a \phi_g^a + \dot{m}^d \bar{A}_\Gamma^a \Delta t + D_v^a \Delta t. \quad (92)$$

Combining the equations outlined in this section results in an implicit model $\mathbf{f}$ for the state of a liquid/gas system undergoing evaporation or condensation. The model is defined by

$$\mathbf{f}(\dot{m}^d, Y_\Gamma^d, T_\Gamma^d, \phi_l^d, \phi_g^d, Y^d, \rho_g^d, E_l^d, E_g^d) = \begin{pmatrix} \dot{m}^d - \rho_g^d \mathcal{D} (Y_\Gamma^d - Y_g^d) (1 - Y_\Gamma^d)^{-1} \Delta_{g,\Gamma}^{-1} \\ Y_\Gamma^d - \mathcal{Y}(T_\Gamma^d, \rho_g^d) \\ T_\Gamma^d + (\dot{H}^d \dot{m}^d - k_g T_g^d \Delta_{g,\Gamma}^{-1} - k_l T_l^d \Delta_{l,\Gamma}^{-1}) / (k_g \Delta_{g,\Gamma}^{-1} + k_l \Delta_{l,\Gamma}^{-1}) \\ \phi_l^d - (\phi_l^a \rho_l^a - \dot{m}^d \bar{A}_\Gamma^a \Delta t) / \rho_l^a \\ \phi_g^d - (1 - \phi_l^d) \\ Y^d - (\rho_g^a Y_g^a \phi_g^a + \dot{m}^d \bar{A}_\Gamma^a \Delta t + D_v^a \Delta t) / (\rho_g^d \phi_g^d) \\ \rho_g^d - (\rho_g^a \phi_g^a + \dot{m}^d \bar{A}_\Gamma^a \Delta t) / \phi_g^d \\ E_l^d + (-\rho_l^a E_l^{d*} \phi_l^a + \mathcal{E}_l^d \dot{m}^d \bar{A}_\Gamma^a \Delta t - k_l (T_\Gamma^d - T_l^d) \bar{A}_\Gamma^a \Delta_{l,\Gamma}^{-1} \Delta t - D_l^a \Delta t) / (\rho_l^d \phi_l^d) \\ E_g^d + (-\rho_g^a E_g^{d*} \phi_g^a - \mathcal{E}_v^d \dot{m}^d \bar{A}_\Gamma^a \Delta t - k_g (T_\Gamma^d - T_g^d) \bar{A}_\Gamma^a \Delta_{g,\Gamma}^{-1} \Delta t - D_g^a \Delta t) / (\rho_g^d \phi_g^d) \end{pmatrix}. \quad (93)$$

Thermodynamic quantities in $\mathbf{f}$ are defined at the advanced time level d , while geometric quantities are defined at the initial time level a . The implicit system is localized, in the sense that all implicit terms are contained within the local computational grid cell: diffusion between computational cells is performed explicitly, while diffusion and phase change within a cell are implicit. This formulation eliminates the need to solve a global problem for diffusion and phase change, while discretely enforcing consistency between phase change, thermal flux, and mass flux. The scheme is conservative with respect to mass, momentum, and energy. Solving system (93) completes the diffusion and phase change procedure, advancing the solution to the post-diffusion time level d . In the present work, system (93) is solved with Newton's method. When possible, components of the Jacobian matrix are evaluated analytically. The fluids considered may obey complex definitions for properties (tabulated properties, for example), requiring some of the Jacobian entries to be evaluated by finite difference.

In practical problems, the situation may arise where the flow is under-resolved, or a computational cell contains a very small volume fraction of liquid or gas ($\phi_l^a = 10^{-8}$, for example). In this case, the default approach to solving the phase change system Eq. (93) may fail to converge, either due to convergence criteria or because a solution does not exist. In the event of nonconvergence, Appendix A provides a secondary lumped model for conservative phase change. The frequency of nonconvergence is considered in subsequent sections, and will be shown to be small.

4.5. Pressure correction

The velocity and total energy are advanced from the diffusion time level d to the advanced time level $n+1$ through a pressure projection defined by

$$\mathbf{u}^{n+1} = \mathbf{u}^d - \Delta t \frac{\nabla p^{n+1}}{\rho^{n+1}}, \quad (94)$$

and

$$E^{n+1} = E^d - \Delta t \frac{\nabla \cdot (\mathbf{u}^{n+1} p^{n+1})}{\rho^{n+1}}, \quad (95)$$

where ρ^{n+1} is the cell-averaged density. In the absence of phase change, the asymptotically preserving formulation of Jemison et al. [36] defines the time-advanced pressure for multiphase flows:

$$\frac{p^{n+1} - p^d}{\rho^{n+1} (c^d)^2 \Delta t^2} - \nabla \cdot \left(\frac{\nabla p^{n+1}}{\rho^{n+1}} \right) = - \frac{\nabla \cdot \mathbf{u}^d}{\Delta t}. \quad (96)$$

The advective pressure p^d is defined by the equation of state of the volume-majority component of each cell, and the speed of sound c^d is also defined by the properties of the volume-majority component.

Phase change introduces the additional complication of a velocity discontinuity at the interface, defined by Eq. (18). Applying this divergence directly would impose an unphysically large momentum increment: because the single-fluid velocity is a mass-averaged quantity applicable to the liquid and the gas, the imposed divergence would accelerate both the liquid and the gas contained in the corresponding face-centered control volume. Rather than directly applying the divergence associated with phase change to the single-fluid velocity, a density correction factor $\mathcal{R}$ is used to apply the appropriate momentum increment, and the phase-specific velocities are corrected after the pressure solve (according to the process to be described in Section 4.5.1). Two assumptions are applied by this approach: first, it is assumed that fluid elements transition phase between the post-advection densities $\rho_{l,i}^a$ and $\rho_{g,i}^a$; second, it is assumed that the appropriate momentum increment for the cell is given by the product of the gas phase density and the velocity change associated with the applied divergence. We therefore define the density correction factor $\mathcal{R}_i$ as ratio of the gas phase density $\rho_{g,i}^{n+1}$ to the minimum adjacent face-centered density,

$$\mathcal{R}_i = \min \left\{ 1.0, \frac{\rho_{g,i}^{n+1}}{\min\{\rho_{i-1/2}^{n+1}, \rho_{i+1/2}^{n+1}\}} \right\}. \quad (97)$$

Under these assumptions, the velocity increment resulting from Stefan flow is applied through an increment to the post-diffusion velocity divergence according to

$$\nabla \cdot \mathbf{u}^s = \nabla \cdot \mathbf{u}^d + \left[\dot{m}^d \left(\frac{1}{\rho_g^a} - \frac{1}{\rho_l^a} \right) \right] \mathcal{R}. \quad (98)$$

The left-hand-side expression $\nabla \cdot \mathbf{u}^s$ quantifies the divergence after the application of Stefan flow, and $\nabla \cdot \mathbf{u}^d$ is the divergence at the post-diffusion state. The second term on the right-hand-side is the divergence resulting from the assumed Stefan flow, multiplied by the density correction factor defined to impose the desired single-fluid momentum change.

Reformulating the pressure equation, Eq. (96), with the appropriate divergence, $\nabla \cdot \mathbf{u}^s$, returns

$$\frac{p^{n+1} - p^d}{\rho^{n+1} (c^d)^2 \Delta t^2} - \nabla \cdot \left(\frac{\nabla p^{n+1}}{\rho^{n+1}} \right) = - \frac{\nabla \cdot \mathbf{u}^d}{\Delta t} - \frac{\dot{m}^d}{\Delta t} \left(\frac{1}{\rho_g^a} - \frac{1}{\rho_l^a} \right) \mathcal{R}. \quad (99)$$

The discretized form of Eq. (99) involves a mixture of cell-centered and face-centered quantities, expressed in one-dimension for cell i by

$$p_i^{n+1} - \rho_i^{n+1} (c_i^d)^2 \Delta t^2 \left(\frac{\left(\frac{p_{i+1}^{n+1} - p_i^{n+1}}{\rho_{i+1/2}^{n+1} \Delta x} - \frac{p_i^{n+1} - p_{i-1}^{n+1}}{\rho_{i-1/2}^{n+1} \Delta x} \right)}{\Delta x} \right) = p_i^d - \rho_i^{n+1} (c^d)_i \Delta t \left[\frac{u_{i+1/2}^d - u_{i-1/2}^d}{\Delta x} + \dot{m}_i^d \left(\frac{1}{\rho_{g,i}^a} - \frac{1}{\rho_{l,i}^a} \right) \mathcal{R}_i \right]. \quad (100)$$

Solving Eq. (100) returns the cell-centered, time-advanced pressure p_i^{n+1} . The Ghost Fluid Method is used to apply the interfacial pressure jump defined by Eq. (19) directly in Eq. (100) [36].

The face-centered, single-fluid velocity is advanced to time level $n+1$ by direct application of Eq. (94), expressed for face $i+1/2$ by

$$u_{i+1/2}^{n+1} = u_{i+1/2}^d - \Delta t \frac{p_{i+1}^{n+1} - p_i^{n+1}}{\rho_{i+1/2}^{n+1} \Delta x}. \quad (101)$$

The face-centered pressure appropriate for projecting the energy has been shown by [37] and [36] to equal

$$p_{i+1/2}^{n+1} = \frac{\rho_{R,i}^{n+1} p_{i+1}^{n+1} + \rho_{L,i+1}^{n+1} p_i^{n+1}}{\rho_{R,i}^{n+1} + \rho_{L,i+1}^{n+1}}, \quad (102)$$

which closes the time-advanced total energy for both phases m according to

$$E_{m,i}^{n+1} = E_{m,i}^d - \Delta t \frac{p_{i+1/2}^{n+1} u_{i+1/2}^{n+1} - p_{i-1/2}^{n+1} u_{i-1/2}^{n+1}}{\rho_i^{n+1} \Delta x}. \quad (103)$$

This work does not update the cell-centered velocity via projection, as in prior work. Instead, the face-centered momentum is distributed to the cell-centers, and the definition of the single-fluid momentum and the Stefan relation are used to compute the phase-specific velocities.

4.5.1. Cell-centered, phase-specific velocities

The single-fluid momentum at time level $n + 1$ is defined as a function of the face-centered velocities by

$$u_i^{n+1} = \frac{u_{i-1/2}^{n+1} \rho_{L,i}^{n+1} |\Omega_{L,i}| + u_{i+1/2}^{n+1} \rho_{R,i}^{n+1} |\Omega_{R,i}|}{\rho_i^{n+1} |\Omega_i|}. \quad (104)$$

The notation used in Eq. (104) for cell i is the same as in Section 4.3.2: the first term in the numerator is the momentum of the left-half of the cell, the second term is the momentum of the right-half of the cell, and the denominator is the cell mass. Equation (104) defines the cell-centered momentum for single-phase cells at time level $n + 1$. Multiphase cells require a method to extract the phase-specific velocities from the known single-fluid momentums.

From the definition of the single-fluid momentum (considering the x -component only), the phase-specific velocities in cell i are assumed to satisfy

$$\phi_{g,i}^{n+1} \rho_{g,i}^{n+1} u_{g,i}^{n+1} + \phi_{l,i}^{n+1} \rho_{l,i}^{n+1} u_{l,i}^{n+1} = \rho_i^{n+1} u_i^{n+1}. \quad (105)$$

From the assumed Stefan flow, the velocity difference between the liquid and gas phases must also satisfy

$$u_{g,i}^{n+1} - u_{l,i}^{n+1} = \dot{m}_i^d \left(\frac{1}{\rho_{g,i}^a} - \frac{1}{\rho_{l,i}^a} \right) n_{x,i}^d, \quad (106)$$

where $n_{x,i}^d$ is the x -component of the reconstructed interfacial normal vector in cell i . Solving Eq. (106) for the gas phase velocity, substituting the result into Eq. (105), and solving for the liquid phase velocity returns

$$u_{l,i}^{n+1} = \frac{\rho_i^{n+1} u_i^{n+1} - \phi_{g,i}^{n+1} \rho_{g,i}^{n+1} \left(\dot{m}_i^d \left(\frac{1}{\rho_{g,i}^a} - \frac{1}{\rho_{l,i}^a} \right) n_{x,i}^d \right)}{\phi_{g,i}^{n+1} \rho_{g,i}^{n+1} + \phi_{l,i}^{n+1} \rho_{l,i}^{n+1}}. \quad (107)$$

The gas phase velocity is then given by rearranging Eq. (106),

$$u_{g,i}^{n+1} = u_{l,i}^{n+1} + \dot{m}_i^d \left(\frac{1}{\rho_{g,i}^a} - \frac{1}{\rho_{l,i}^a} \right) n_{x,i}^d. \quad (108)$$

Extracting the phase-specific velocities from the single-fluid momentum is effectively a liquid/gas momentum transport process, anticipated by including the density correction factor in Eq. (98). The change in gas phase momentum in cell i associated with extracting the phase-specific velocity from the single-fluid velocity is given by

$$\delta P_{g,i} = (u_{g,i}^{n+1} - u_i^{n+1}) \rho_{g,i}^{n+1} \phi_{g,i}^{n+1} V, \quad (109)$$

with associated velocity change

$$\delta u_{g,i} = u_{g,i}^{n+1} - u_i^{n+1}. \quad (110)$$

In order to decrease sensitivity of the post-projection solution to the local volume fraction, we expand this liquid/gas momentum exchange to occur not only within the interfacial cell i , but also between the adjacent face-centered control volumes $i - 1/2$ and $i + 1/2$, provided the following conditions are satisfied: (1) cell i is an interfacial cell; (2) cell $i + 1$ is a pure gas phase cell, and cell $i - 1$ is a pure liquid phase cell (or vice versa). To simplify the following description, we assume a planar interface where the liquid is on the left and the gas is on the right.

We assume a momentum exchange between the gas in cell $i + 1$ and the liquid in cell $i - 1$, characterized by the velocity increment $\delta u_{g,i}$, mass $\rho_{g,i+1}^{n+1} |\Omega_{L,i+1}|$, and the density correction factor $\mathcal{R}_i$:

$$u_{g,i+1}^{n+1} = \frac{\rho_{g,i+1}^{n+1} u_{i+1}^{n+1} V + \delta u_{g,i} \rho_{g,i+1}^{n+1} |\Omega_{L,i+1}| (1.0 - \mathcal{R}_i)}{\rho_{g,i+1}^{n+1} V} \quad (111)$$

and

$$u_{l,i-1}^{n+1} = \frac{\rho_{l,i-1}^{n+1} u_{i-1}^* V - \delta u_{g,i} \rho_{g,i+1}^{n+1} |\Omega_{L,i+1}| (1.0 - \mathcal{R}_i)}{\rho_{l,i-1}^{n+1} V}. \quad (112)$$

The superscript $*$ denotes the single-fluid velocity resulting from Eq. (104), which would typically be considered the time $n + 1$ velocity, but has undergone an additional modification to finally attain the time level $n + 1$ (and thus is temporarily denoted by $*$). We note the momentum increment added to the gas phase in cell $i + 1$ in Eq. (111) is subtracted from the liquid phase in cell $i - 1$ in Eq. (112), preserving conservation. Face-centered momentum exchange resulting from the phase-specific velocity extraction is applied to face $i + 1/2$ by

$$u_{i+1/2}^{n+1} = \frac{M_{i+1/2}^{n+1} u_{i+1/2}^* + \left(\delta u_{g,i} \rho_{g,i+1}^{n+1} |\Omega_{L,i+1}| (1.0 - \mathcal{R}_i) + \delta P_{g,i} \right)}{M_{i+1/2}^{n+1}}, \quad (113)$$

and to face $i - 1/2$ by

$$u_{i-1/2}^{n+1} = \frac{M_{i-1/2}^{n+1} u_{i-1/2}^* - \left(\delta u_{g,i} \rho_{g,i+1}^{n+1} |\Omega_{L,i+1}| (1.0 - \mathcal{R}_i) + \delta P_{g,i} \right)}{M_{i-1/2}^{n+1}}. \quad (114)$$

Analogous expressions result from considering the liquid on the right and the gas on the left. Kinetic energy transfer via this mechanism is neglected.

4.5.2. Face-centered, phase-specific velocities

The face-centered, phase-specific velocities serve as the advective velocities, $u_{a,l}$ and $u_{a,g}$, defined in Section 4.3. Far from the liquid/gas interface (in single-phase regions), the face-centered velocity is defined by Eq. (101). In face-centered control volumes containing both phases, the single-fluid velocities are extracted from the face-centered, single-fluid momentum using a procedure similar to the cell-centered case described in Section 4.5.1.

The liquid/gas advective velocity difference is defined by

$$\delta u_a = u_{a,g} - u_{a,l}. \quad (115)$$

For a multiphase face-centered control volume $i + 1/2$, the liquid/gas advective velocity difference is chosen according to

$$\delta u_{a,i+1/2} = \begin{cases} \dot{m}_i^d \left(\frac{1}{\rho_{g,i}^d} - \frac{1}{\rho_{l,i}^d} \right) n_{x,i}^d & \text{if } \phi_{g,i} < \phi_{g,i+1} \text{ and } \phi_{g,i} > 0 \\ \dot{m}_{i+1}^d \left(\frac{1}{\rho_{g,i+1}^d} - \frac{1}{\rho_{l,i+1}^d} \right) n_{x,i+1}^d & \text{else.} \end{cases} \quad (116)$$

The liquid advective velocity is extracted from the face-centered single-fluid momentum according to

$$u_{a,l,i+1/2}^{n+1} = \frac{M_{i+1/2}^{n+1} u_{i+1/2}^{n+1} - M_{g,i+1/2}^{n+1} \delta u_{a,i+1/2}}{M_{i+1/2}^{n+1}}, \quad (117)$$

where $M_{g,i+1/2}^{n+1}$ is the gas phase mass contained in $i + 1/2$. The gas advective velocity is then given by

$$u_{a,g,i+1/2}^{n+1} = u_{a,l,i+1/2}^{n+1} + \delta u_{a,i+1/2}. \quad (118)$$

A simple strategy is applied to extend the phase-specific advective velocities into the unlike phase. For both phases m , constant extrapolation is used to populate advective velocities in face-centered control volumes that contain zero phase m mass. Two additional procedures are applied in the liquid phase only. Extending the notation of Section 4.3.2, we define left-half and right-half volume fractions of the liquid phase in cell i by

$$L_i = \frac{2}{V} \int_{\Omega_{L,i}} \chi_l d\Omega, \quad (119)$$

and

$$R_i = \frac{2}{V} \int_{\Omega_{R,i}} \chi_l d\Omega, \quad (120)$$

respectively. Using this notation, a second extrapolation procedure is applied to the liquid advective velocity:

$$u_{a,l,i+1/2}^{n+1} = \begin{cases} u_{a,l,i+3/2}^{n+1} & \text{if } 0.1 > R_i^{n+1} < L_{i+1}^{n+1} < 1/2 \text{ and } L_{i+1}^{n+1} < R_{i+1}^{n+1} \text{ and } \phi_{l,i-1}^{n+1} = 0; \\ u_{a,l,i-1/2}^{n+1} & \text{if } 0.1 > L_i^{n+1} < R_{i-1}^{n+1} < 1/2 \text{ and } R_{i-1}^{n+1} < L_{i-1}^{n+1} \text{ and } \phi_{l,i+1}^{n+1} = 0; \\ u_{a,l,i+1/2}^{n+1} & \text{else.} \end{cases} \quad (121)$$

Finally, the liquid advective velocity in the region near the interface is defined by a kernel approximation of the cell-centered liquid phase velocities. The kernel approximation is applied if both adjacent cell-centered control volumes contain less than 5% liquid volume fraction. For cell faces that satisfy these criteria, the kernel approximation is defined by

$$u_{a,l,i+1/2}^{n+1} = \frac{\sum_j u_{l,j}^{n+1} W_{ij}}{\sum_j W_{ij}}. \quad (122)$$

The weight function W_{ij} is given by

$$W_{ij} = \begin{cases} \left(1 - \left(\frac{x_{i+1/2} - X_{l,j}^{n+1}}{r_e}\right)^2\right)^4 (\phi_{l,j}^{n+1}), & \frac{\|x_{i+1/2} - X_{l,j}^{n+1}\|}{r_e} < 1; \\ 0, & \frac{\|x_{i+1/2} - X_{l,j}^{n+1}\|}{r_e} \geq 1, \end{cases} \quad (123)$$

where $r_e = 2.9\Delta x$ is the non-zero influence radius, $x_{i+1/2}$ is the centroid of the cell face, and $X_{l,j}^{n+1}$ is the liquid phase center of mass in neighboring cell j . The form of W_{ij} and the span of the influence radius r_e were identified via empirical testing, and optimization is a potential area for future work.

4.6. Auxiliary operations

Additional procedures are required to achieve conservation and robustness. These procedures are included in this section because including them in the earlier sections would obfuscate the description of the method, and because they can be modified without fundamental changes to the approach.

4.6.1. In-cell pressure correction

Velocity divergence in a mixed-phase cell modifies the density of both phases within the cell, resulting in different liquid phase and gas phase pressures. An in-cell pressure correction is used to reinforce the required jump-condition in pressure at the interface, given by Eq. (19). The pressure correction is performed by conservatively modifying the densities and VOFs within the computational cell.

The pressure correction procedure is executed before and after solving the implicit phase change system, Eq. (93). The starting values of liquid density and VOF are defined by ρ_{ls} and ϕ_{ls} . The gas density and VOF are analogously defined by ρ_{gs} and ϕ_{gs} . A bisection method is used to iteratively find the solution of the following system of equations:

$$p_l(T_l, \rho_{lf}) - \left[p_g(T_g, \rho_{gf}) + \kappa\sigma + \dot{m}^2 \left(\frac{1}{\rho_{gf}^a} - \frac{1}{\rho_{lf}^a} \right) \right] = 0, \quad (124)$$

$$\rho_{lf} = \frac{\rho_{li}\phi_{li}}{\phi_{li} + \delta\phi}, \quad (125)$$

$$\phi_{lf} = \phi_{li} + \delta\phi, \quad (126)$$

$$\rho_{gf} = \frac{\rho_{gi}\phi_{gi}}{\phi_{gi} - \delta\phi}, \quad (127)$$

$$\phi_{gf} = \phi_{gi} - \delta\phi, \quad (128)$$

where subscripts lf and gf denote the liquid/gas density and VOF values that satisfy the jump condition in pressure, defined by Eq. (124). Consistency between the liquid and gas phase pressures within the interfacial cell is therefore restored by perturbing the liquid phase volume fraction by $\delta\phi$, and the gas phase volume fraction by $-\delta\phi$, such that the jump condition for pressure, Eq. (124), is satisfied within the cell.

4.6.2. Reference center of mass update

The reference center of mass $\mathbf{X}_{\text{Ref}}$ advances from state $\mathbf{X}_{\text{Ref}}^a$ to $\mathbf{X}_{\text{Ref}}^d$ in order to remain consistent with the VOF. Subscripts specifying the liquid and gas phases do not appear in this section, as the procedure is identical in both cases.

Referencing the minimization given by Eq. (27), the reconstructed interface at time level a is defined by

$$(\mathbf{n}^a, b^a) = \underset{\mathbf{n}^a, b^a, \phi_{\text{Ref}}^a = \phi_{\text{Recon}}^a}{\operatorname{argmin}} \quad \|\mathbf{X}_{\text{Ref}}^a - \mathbf{X}_{\text{Recon}}^a\|. \quad (129)$$

The reconstructed normal vector is assumed to remain unchanged from time level a to time level d , defining the reconstruction at time level d according to

$$\mathbf{n}^a \cdot (\mathbf{x} - \mathbf{x}_i) + b^d = 0, \quad (130)$$

constrained by $\phi^d = \phi_{\text{Recon}}^d(\mathbf{n}^a, b^d)$. Further, the displacement of the reconstructed center of mass is assumed equal to the displacement of the reference center of mass, allowing for advancement of the reference center of mass to time level d according to

$$\mathbf{x}_{\text{Ref}}^d = \mathbf{x}_{\text{Ref}}^a + (\mathbf{x}_{\text{Recon}}^d - \mathbf{x}_{\text{Recon}}^a). \quad (131)$$

4.7. Post-diffusion, face-centered momentum update

Small motions of the phase interface due to phase change alter the face-centered momentum and require special treatment to maintain conservation. The post-advection, face-centered momentum is defined by

$$P_{i+1/2}^a = u_{i+1/2}^a M_{i+1/2}^a. \quad (132)$$

Motion of the interface due to evaporation changes the face-centered mass $M_{i+1/2}^a$, such that $M_{i+1/2}^a \neq M_{i+1/2}^d$. In this work, we simply assume the face-centered momentums to be unaffected by the motion of the interface, and modify the face centered velocity according to

$$u_{i+1/2}^* = u_{i+1/2}^a \left(\frac{M_{i+1/2}^a}{M_{i+1/2}^d} \right). \quad (133)$$

This procedure ensures conservation of face-centered and cell-centered momentum through the phase change and diffusion procedure, despite the motion of the interface.

After correcting the face-centered velocities, viscous effects must also be applied to the face-centered velocity prior to solving the Helmholtz equation for pressure. We define the viscous forcing term computed at the cell-center and applied to the cell-centered velocity in Eq. (72) by

$$\mathbf{a}_{\mu,i} = (\nabla \cdot \boldsymbol{\tau}^a)_i \Delta t. \quad (134)$$

The viscous forcing term is conservatively distributed from the cell-centers to the cell-faces, returning the post-diffusion, face-centered velocities,

$$u_{i+1/2}^d = u_{i+1/2}^* + \frac{a_{\mu,i-1} \rho_{R,i-1}^d + a_{\mu,i} \rho_{L,i}^d}{\rho_{R,i-1}^d + \rho_{L,i}^d}, \quad (135)$$

where $a_{\mu,i-1}$ is the x -component of $\mathbf{a}_{\mu,i-1}$.

4.8. Phase-specific advective velocity augmentation

The interfacial mass fraction depends on the gas density according to Eq. (79). Reductions in interfacial density result in increased interface mass fraction, potentially leading to further reductions in density. To increase robustness, the interfacial velocity difference between the liquid and gas is modified for the first step of the fixed-point iteration. The modification is based upon a comparison between the gas density in cell i , and a representative density of the gas in the immediately adjacent cells j , characterized here by a filtered density,

$$\overline{\rho_{g,i}^a} = \frac{\sum_j \phi_{g,j}^a \rho_{g,j}^a T_{g,i}^a / T_{g,j}^a}{\sum_j \phi_{g,j}^a}. \quad (136)$$

To mitigate density variations attributable to temperature gradients near the interface, the temperature ratio, $T_{g,i}^a / T_{g,j}^a$, augments the cell j density (this assumes pressure variations are much smaller than temperature and density variations between i and j). In the case of evaporation, if the density $\rho_{g,i}^a$ is low relative to the filtered density $\overline{\rho_{g,i}^a}$, the liquid/gas velocity difference should be decreased (thereby increasing $\rho_{g,i}^a$). Conversely, if the density is high, the velocity difference should be increased (thereby decreasing $\rho_{g,i}^a$). Condensation requires a correction with the inverse relationship to density. A correction strategy of this type is achieved by defining a multiplier $\mathcal{C}$,

$$\mathcal{C}_i = \begin{cases} \min \left\{ 2.0, \left(\rho_{g,i}^a / \overline{\rho_{g,i}^a} \right)^8 \right\} & \dot{m}_i^d > 0.0; \\ \min \left\{ 2.0, \left(\overline{\rho_{g,i}^a} / \rho_{g,i}^a \right)^8 \right\} & \dot{m}_i^d < 0.0. \end{cases} \quad (137)$$

In the first pass of the fixed-point iteration, the phase change rate $\dot{m}_i^d$ present in Sections 4.5.1 and 4.5.2 is multiplied by the correction factor C_i – these sections pertain to extracting phase-specific velocities from single-fluid velocities. The liquid/gas velocity difference applied in the first pass of the fixed-point iteration is therefore defined by

$$u_{g,i}^{n+1} - u_{l,i}^{n+1} = \left[\dot{m}_i^d \left(\frac{1}{\rho_{g,i}^a} - \frac{1}{\rho_{l,i}^a} \right) n_{x,i}^d \right] C_i. \quad (138)$$

The second pass of the fixed-point iteration does not include this correction procedure, which is equivalent to setting $C_i = 1$.

A final modification of the phase-specific advective velocities considers mass/volume inconsistencies that may have occurred during the advection procedure, defined on lines 17 and 22 of Algorithm 2. In order to prevent repeated application of the consistency restoring procedures defined on lines 17-26 of Algorithm 2, the following two-step operation is performed:

1. If a computational cell i executes a consistency restoring procedure during advection (lines 17-26 of Algorithm 2), the cell is flagged.
2. After defining a new set of time-advanced advective velocities (either $\mathbf{u}_{a,l}^*$ and $\mathbf{u}_{a,g}^*$ or $\mathbf{u}_{a,l}^{n+1}$ and $\mathbf{u}_{a,g}^{n+1}$), a search is performed for cells that have been flagged in step 1. If a flagged cell i is found, the volume-majority phase in cell i is compared to the adjacent cells (the cells that share a face with cell i). If the volume-majority phase is not the same in cell i and an adjacent cell, the gas phase advective velocity at the shared cell face is set equal to the co-located liquid phase advective velocity.

This procedure provides a small perturbation to the solution, and was found to prevent locally recurring mass/volume inconsistency.

5. Results

Verification of the framework is performed in a progressive manner. The Sod-shock test problem is examined first to verify the description of single phase compressible flow. Next, the time-splitting and sharp-interface discretization of diffusion is verified by considering transient, one-dimensional, interfacial heat transfer. Thermal and species transport mechanisms are then combined with the evaporation dynamic to verify evaporation of a one-dimensional film. The final set of verification tests considers the evaporation rate of three-dimensional droplets in stationary and translating configurations. Following verification, validation is performed via comparison to an empirical correlation for the evaporation rate of droplets in cross-flow. The section concludes with the simulation of a high pressure iso-octane spray.

5.1. Sod-shock tube

Prior work of Kwatra et al. [37] and Jemison et al. [36] used the Sod-shock test case to verify their asymptotically preserving compressible flow schemes. This verification test is repeated for the present methodology.

Problem setup The Sod-shock test problem is performed in a one-dimensional domain with extent $0 \leq x \leq 1$. Initial conditions for density, pressure, and velocity are given by

$$\rho_0 = \begin{cases} 1 & x \leq 0.5; \\ 0.125 & x > 0.5, \end{cases} \quad (139)$$

$$p_0 = \begin{cases} 1 & x \leq 0.5; \\ 0.1 & x > 0.5. \end{cases} \quad (140)$$

and

$$u_0 = 0. \quad (141)$$

The fluid is an ideal gas with $\gamma = 1.4$ and an equation of state given by

$$p = (\gamma - 1) \rho e, \quad (142)$$

where e is the specific internal energy. The maximum stable time step for this configuration is given by

$$\Delta t_{max} = \frac{1}{N_x}, \quad (143)$$

where N_x is the number of computational nodes in the x -direction.

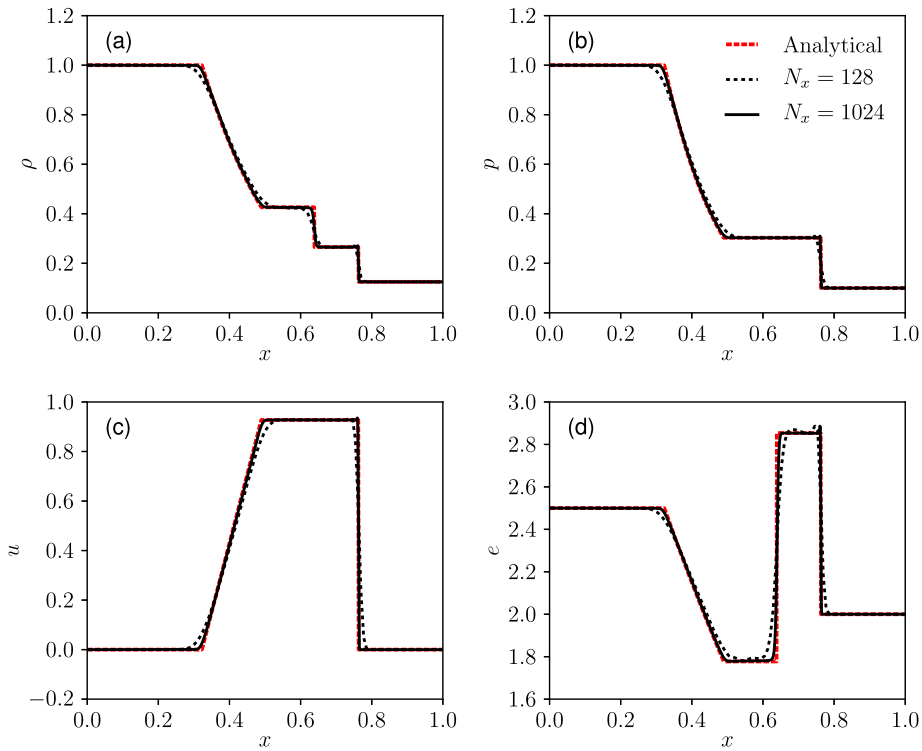


Fig. 6. Analytical solution to the Sod-shock problem at $t = 0.15$ (solid red) compared with simulation results for mesh sizes $N_x = 128$ (dotted black) and $N_x = 1024$ (solid black): (a) density; (b) pressure; (c); velocity; (d) internal energy.

Simulation errors are quantified via the L_1 error evaluated relative to the analytical solution. The L_1 error in fluid property η is defined by

$$L_1 = \int_0^1 |\eta_s(x, t_1) - \eta(x, t_1)| dx, \quad (144)$$

where $\eta_s(x, t_1)$ is the simulated property at position x and time t_1 , and $\eta(x, t_1)$ is the corresponding analytical quantity.

Numerical specification The Sod shock problem is simulated at five mesh resolutions: $N_x = 64$, $N_x = 128$, $N_x = 256$, $N_x = 512$, and $N_x = 1024$. Each simulation is performed in a pseudo one-dimensional configuration, where $N_y = N_z = 4$ mesh spacings are considered in the y - and z -directions. The time step is defined by $\Delta t = 0.25 \times \Delta t_{max}$ for every mesh resolution considered.

Results The analytical solution to the Sod shock problem is shown in Fig. 6 at time $t = 0.15$, along with the simulation results produced by the $N_x = 128$ and $N_x = 1024$ simulations. Panels (a)-(d) show the density, pressure, velocity, and internal energy, respectively. The $N_x = 128$ mesh produces dissipative errors in all properties, in addition to dispersive errors in the internal energy between $x = 0.6$ and $x = 0.8$. Increasing the resolution to $N_x = 1024$ decreases dissipative error throughout the domain, although dispersive error remains near the right-most discontinuity in internal energy. The results presented in Fig. 6 are consistent with other asymptotically preserving schemes [37,36].

Fig. 7 provides the L_1 errors as a function of mesh resolution N_x . The dotted line indicates convergence at a rate with $N_x^{-1/2}$ and the dashed line indicates convergence with N_x^{-1} . Density, pressure, velocity, and internal energy all converge at a rate between half and first-order with respect to increasing mesh resolution over the range of resolutions considered. These convergence rates are consistent with those presented by Jemison et al. [36].

5.2. One-dimensional heat transfer

One-dimensional transient heat conduction is considered between two semi-infinite domains joined at an interface, where the left half-plane is comprised of cool air and the right half-plane of warm water. This problem has been used previously for verification of interfacial heat transfer methods [41].

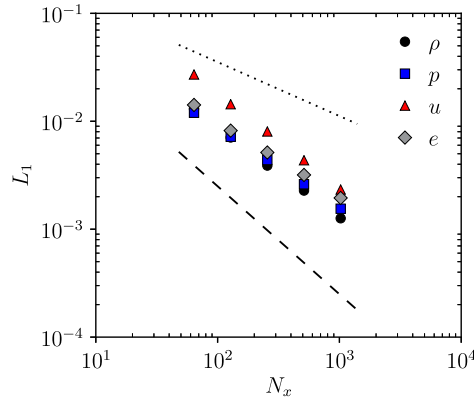


Fig. 7. L_1 property errors for the Sod-shock test problem as a function of mesh resolutions ranging from $N_x = 64$ to $N_x = 1024$. The dotted line illustrates convergence with $N_x^{-1/2}$, and the dashed line denotes convergence with first-order.

Table 1

Fluid properties for one-dimensional heat transfer verification.

Property	Water	Air
ρ [g/cm ³]	0.999	0.003
k [$\frac{\text{erg/s}}{\text{cm K}}$]	5.8801×10^4	2.587×10^3
c_v [erg/g K]	4.1855×10^7	7.71725×10^6
α [cm ² /s]	1.40628×10^{-3}	1.11741×10^{-1}

Problem setup The problem domain spans from $x = -L$ to $x = L$ with the interface located at $x = 0 + \epsilon$. The half-length of the simulation domain is $L = 0.02$ cm. The perturbation ϵ is a small number (smaller than the mesh spacing Δ_x) used to translate the interface relative to the mesh as a means to investigate performance sensitivity to the interface location within an interfacial cell. At time $t = 0$, the air has a temperature of $T = 300$ K and the water has a temperature of $T = 350$ K. As time advances, energy diffuses from the water into the air, increasing the air temperature and decreasing the water temperature. The analytical solution to the problem is given by [41] (for $\epsilon = 0$)

$$T(x, t) = \begin{cases} T_g + T_c \times \text{Erfc}(-x / (2\sqrt{\alpha_g t})) & x < 0 \\ T_g + T_c \times \left(1 + \frac{k_g}{k_l} \sqrt{\frac{\alpha_l}{\alpha_g}} \text{Erf}(x / (2\sqrt{\alpha_g t}))\right) & x \geq 0 \end{cases} \quad (145)$$

where

$$T_c = (T_l - T_g) \frac{k_l / \sqrt{\alpha_g}}{k_l / \sqrt{\alpha_l} + k_g / \sqrt{\alpha_g}}. \quad (146)$$

The constant thermal diffusivity $\alpha = k / \rho c_v$, thermal conductivity k , specific heat c_v , and density ρ are provided for the liquid and gas phases in Table 1. The thermal diffusivity of air is two orders of magnitude greater than water (a magnitude of 0.1 cm²/s for air and 0.001 cm²/s for water). The solution is therefore strongly dependent on which side of the interface is considered, with greater temperature change expected in the air than in the water.

To evaluate the performance of the sharp-interface diffusion model, we consider the dependence of the numerical solution on the mesh resolution N_x , the position of the interface in the interfacial Eulerian cell, and the time step Δt . In each evaluation, the accuracy of the solution is compared to the analytical solution by means of the L_∞ error, defined by

$$L_\infty = \max \left| \frac{T_s(x, t_1) - T(x, t_1)}{T_l - T_g} \right|, \quad (147)$$

where T_s is the simulated temperature, T is the analytical temperature provided by Eq. (145), and $t_1 = 3 \times 10^{-5}$ s is the evaluation time.

Numerical specification The simulation domain is comprised of N_x computational nodes in the x -direction, and 4 computational nodes in the y - and z -directions (a pseudo one-dimensional implementation). Zero-flux Neumann boundary conditions are implemented on all domain boundaries, such that $\partial T / \partial n = 0$, where n is the boundary-normal direction.

A set of verification cases is constructed by considering the L_∞ error as a function of the mesh resolution N_x , the location of the interface within the interfacial cell (defined by the perturbation ϵ), and the simulation time step size Δt .

Table 2
Simulation case definitions for one-dimensional heat transfer verification.

Case	Mesh size N_x	Perturbation ϵ [cm]	Max time step Δt_{max} [s]
1	16	6.25×10^{-6}	2.6×10^{-5}
2	32	6.25×10^{-6}	6.5×10^{-6}
3	64	6.25×10^{-6}	1.6×10^{-6}
4	128	6.25×10^{-6}	4.1×10^{-7}
5	256	6.25×10^{-6}	1.0×10^{-8}
6	64	-3.125×10^{-4}	1.6×10^{-6}
7	64	-6.25×10^{-6}	1.6×10^{-6}
8	64	0	1.6×10^{-6}
9	64	6.25×10^{-6}	1.6×10^{-6}
10	64	3.125×10^{-4}	1.6×10^{-6}

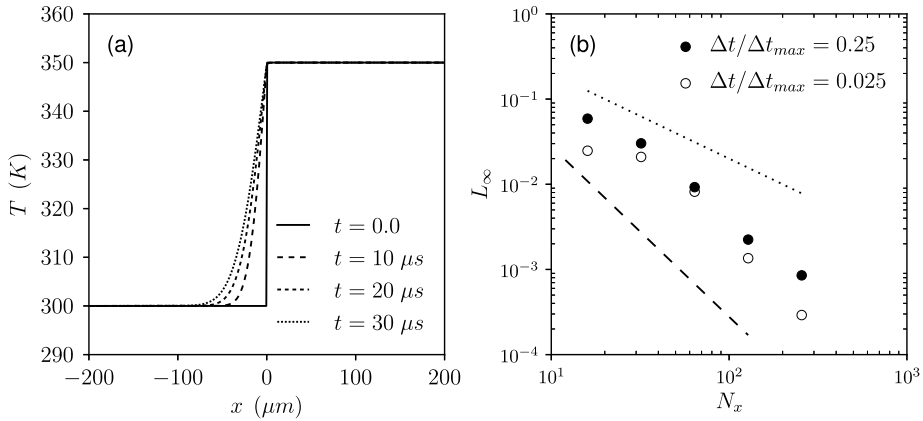


Fig. 8. Performance assessment for one-dimensional heat transfer: (a) analytical solution for temperature in air (left half-plane) and water (right half-plane) at times $t = 0, 10, 20$, and $30 \mu s$; (b) L_∞ error evaluated at time $t_1 = 30 \mu s$ for mesh sizes of $N_x = 16, 32, 64, 128$, and 256 (Cases 1–5) for time steps $\Delta t = 0.25 \times \Delta t_{max}$ and $\Delta t = 0.025 \times \Delta t_{max}$. The dotted line denotes first-order convergence and the dashed line denotes second-order convergence.

Considering each of these parameters results in two sets of cases outlined in Table 2. Cases 1–5 evaluate performance for mesh resolutions spanning from $N_x = 16$ (Case 1) to $N_x = 256$ (Case 5). In Cases 1–5, the time step is defined by $\Delta t = 0.25 \times (2L/N_x)^2 / \alpha_g$ and the interface perturbation $\epsilon = 6.25 \times 10^{-6}$ cm. The small perturbation ϵ moves the interface slightly out of alignment with the mesh, exercising the sharp-interface capability of the method. Cases 6–10 exercise the dependence of performance on the location of the interface within the interfacial cell by varying the perturbation parameter ϵ from $\epsilon = -0.5\Delta x$ to $\epsilon = 0.5\Delta x$ for a fixed mesh size of $N_x = 64$ and a fixed time step of $\Delta t = 0.25 \times \Delta t_{max}$. Finally, the dependence of the solution on time step is considered by repeating Cases 1–10 for a time step reduced by a factor of 10 to $\Delta t = 0.025 \times \Delta t_{max}$.

Results Fig. 8 panel (a) presents the temporal evolution of the analytical solution from the initial condition $t = 0$ s to the evaluation time $t_1 = 3 \times 10^{-5}$ s. Initially, the air to the left of $x = 0$ is uniformly at temperature $T = 300$ K and the water to the right of $x = 0$ is uniformly at temperature $T = 350$ K. By time $t = 1 \times 10^{-5}$ s, the temperature of the air near the interface has increased significantly, producing a smooth transition between the far-field temperatures. The liquid temperature has not visibly changed due to the comparatively high thermal capacitance of water. As time progresses, the temperature of the air continues to rise as energy diffuses away from the interface.

The simulation errors corresponding to Cases 1–5 are presented in Fig. 8 panel (b) for time steps of $\Delta t = 0.25 \times \Delta t_{max}$ and $\Delta t = 0.025 \times \Delta t_{max}$. As the node count increases from $N_x = 16$ to $N_x = 256$, the maximum error decreases from ≈ 0.025 to ≈ 0.00085 for $\Delta t = 0.25 \times \Delta t_{max}$ and from ≈ 0.059 to ≈ 0.00029 for $\Delta t = 0.025 \times \Delta t_{max}$. Less than first-order convergence is observed for mesh resolutions smaller than $N_x = 32$, while approximately second-order convergence is observed over the remaining resolutions considered.

Errors corresponding to Cases 6–10 are presented in Fig. 9, demonstrating the sensitivity of the solution to the position of the interface within the interfacial cell. For each case, the computational mesh contains $N_x = 64$ nodes and considers time steps of $\Delta t = 0.25 \times \Delta t_{max}$ and $\Delta t = 0.025 \times \Delta t_{max}$. For the larger time step of $\Delta t = 0.25 \times \Delta t_{max}$, the maximum error $L_\infty = 0.013$ occurs for $\epsilon/\Delta x = 0.01$ and the minimum error $L_\infty = 0.0068$ for $\epsilon/\Delta x = 0.0$ – approximately a factor of two change in the error across the range of ϵ considered. The smaller time step of $\Delta t = 0.025 \times \Delta t_{max}$ is less sensitive to ϵ , with maximum and minimum errors of $L_\infty = 0.054$ and $L_\infty = 0.43$ corresponding to $\epsilon/\Delta x = 0.01$ and $\epsilon/\Delta x = -0.5$, respectively.

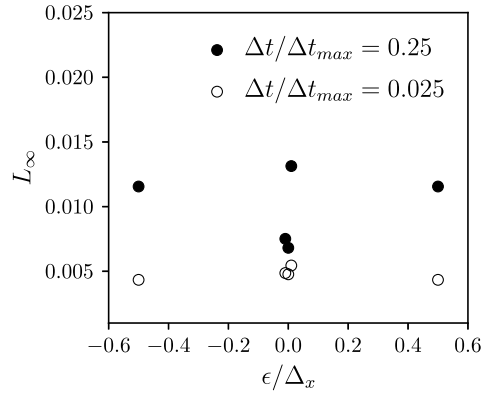


Fig. 9. L_∞ error in the one-dimensional heat transfer simulation evaluated at time $t_1 = 30 \mu\text{s}$ for interface perturbations ϵ ranging from $\epsilon = -0.5\Delta x$ to $\epsilon = 0.5\Delta x$ (Cases 6-10). Results for each perturbation ϵ are considered for a moderate time step of $\Delta t = 0.25 \times \Delta t_{max}$ (solid black circles) and a small time step $\Delta t = 0.025 \times \Delta t_{max}$ (empty circles). All cases run on an $N_x = 64$ mesh.

5.3. Planar evaporation

Evaporation from a flat surface provides the first verification assessment for evaporation. The flat surface provides a simple geometry that allows for inspection of a wide range of evaporation rates.

Problem setup The steady-state evaporation rate from a flat surface can be expressed as a function of the mass transfer conductance, g , and the mass transfer driving force, B_m , according to

$$\dot{m} = gB_m, \quad (148)$$

where $B_m = (Y_\Gamma - Y_\infty)/(1 - Y_\Gamma)$, and Y_∞ is the far-field vapor mass fraction [9]. As the mass transfer driving force approaches zero, the mass transfer conductance approaches a limiting value g^* , given by

$$g^* = \lim_{\dot{m} \rightarrow 0} \frac{\dot{m}}{B_m}. \quad (149)$$

In this limiting regime, the vapor distribution is assumed to be linear over an evaporation film thickness, δ_v , such that

$$g^* = \frac{\rho_g \mathcal{D}}{\delta_v}. \quad (150)$$

In the special case of Couette flow, where the free-stream maintains far-field values at the film-thickness δ_v , the evaporation rate is given by

$$\dot{m} = \frac{\rho_g \mathcal{D}}{\delta_v} \ln \left[1 + \frac{Y_\Gamma - Y_\infty}{1 - Y_\Gamma} \right]. \quad (151)$$

The corresponding surface-normal velocity v is then

$$v = \dot{m} \left[\frac{1}{\rho_g} - \frac{1}{\rho_l} \right]. \quad (152)$$

We use Eq. (151) to verify the evaporation rate for a fixed interface vapor mass fraction.

Numerical specification We consider a pseudo one-dimensional domain with height $H = 0.04 \text{ cm}$ in the y -direction. The domain widths in the x - and z -directions are set to four mesh spacings. The bottom 3/10 of the domain contains liquid and the top 7/10 of the domain contains gas. The lower surface is an insulated slip wall, while the upper surface is an outlet defined by pressure $p_o = 9.13 \times 10^6 \text{ dyne/cm}^2$. The definitions of the verification cases considered are outlined in Table 3. Evaluating sensitivity to mesh resolution, Cases 1-4 consider an interfacial vapor mass fraction of $Y_\Gamma = 0.1$ for mesh sizes of $N_y = 8, 16, 32$, and 64 . Then, considering sensitivity to interfacial vapor mass fraction, Cases 5-7 consider interfacial vapor mass fractions of $Y_\Gamma = 0.3, 0.6$, and 0.9 , for a fixed mesh size of $N_y = 32$.

Temperature, density, and transport coefficients are initialized uniformly within each phase and are defined in Table 4. For the purposes of this evaluation, thermal conductivities, viscosities, and species diffusion coefficients are constrained to the constant values provided in Table 4. The air/vapor mixture is defined as an ideal gas with

$$p_g = (\gamma - 1) \rho c_v T, \quad (153)$$

Table 3
Simulation case definitions for planar evaporation verification.

Case	Mesh size N_y	Interfacial mass fraction Y_Γ	time step Δt [s]
1	8	0.1	3.2×10^{-6}
2	16	0.1	1.6×10^{-6}
3	32	0.1	8.0×10^{-7}
4	64	0.1	4.0×10^{-7}
5	16	0.3	1.6×10^{-6}
6	16	0.6	1.6×10^{-6}
7	16	0.9	1.6×10^{-6}

Table 4
Initial fluid properties for planar evaporation verification.

Property	Water	Air
T_o [K]	362	362
ρ_o [g/cm ³]	0.999	0.00879
k [erg/s cm K]	1.69×10^4	6.316×10^3
μ [g/cm s]	5.8×10^{-3}	1.871×10^{-4}
c_v [erg/g K]	4.1855×10^7	7.71725×10^6
$\mathcal{D}$ [cm ² /s]	0.035	0.035

where $\gamma = 1.4$, and the water is governed by a Tait equation of state,

$$p_l = B \left((\rho/\rho_o)^{7.15} - 1 \right) + p_o, \quad (154)$$

where $B = 3.31 \times 10^9$ dyne/cm². Both fluids are initially quiescent with $v_o = 0$. The vapor mass fraction is initialized with a linear profile given by

$$Y(y, t_o) = 0.9 \left(1 - \frac{(y - y_\Gamma)}{\delta_v(t_o)} \right) Y_\Gamma, \quad (155)$$

where $Y(y, t_o)$ is the initial vapor mass fraction at height y and $\delta_v(t_o) = 7H/10$ is the initial vapor film thickness. The vapor mass fraction is initialized to introduce a relaxation period over which the evaporation rate approaches the analytical value. In order to minimize changes in gas density due to evaporative cooling (the analytical solution assumes constant density), the specific internal energy of water vapor is set equal to the specific internal energy of liquid water for this test.

The film thickness δ_v increases slightly as the liquid evaporates. The error in evaporation rate $E_{\dot{m}}$ is computed as a function of the transient film thickness $\delta_v(t)$ according to

$$E_{\dot{m}}(t) = \left| \frac{\dot{m}_s(t) - \dot{m}}{\dot{m}} \right|, \quad (156)$$

where $\dot{m}_s$ is the simulated evaporation rate, and $\dot{m}$ is the rate provided by Eq. (151). The analytical evaporation rate $\dot{m}$ in Eq. (156) is evaluated as a function of the initial gas density, such that errors in interfacial gas density are captured.

Results Fig. 10 presents evaporation rate as a function of mesh resolution in panel (a). The evaporation rate defined by Eq. (151) appears as a dotted red line for reference. During the relaxation transient ($t < 6$ ms), the evaporation rate exceeds the value provided by Eq. (151) for all mesh resolutions. Only minor variation is apparent among the mesh resolutions, with the most significant deviation presented by $N_y = 8$ at early times. All four resolutions approach the value provided by Eq. (151) for $t > 8$ ms. Fig. 10 panel (b) provides the evaporation rate error $E_{\dot{m}}$ corresponding to the results in panel (a). Errors decrease monotonically for all cases, and errors decrease with increasing mesh resolution for $t > 6$ ms. For all mesh resolutions considered, errors decrease to less than 1% after relaxing from the perturbed initial condition.

Fig. 11 presents evaporation rate as a function of interfacial mass fraction for $N_y = 16$. Panel (a) provides simulation results for $Y_\Gamma = 0.3$, $Y_\Gamma = 0.6$, and $Y_\Gamma = 0.9$, along with the corresponding values given by Eq. (151) as red lines of the same dash/dot configuration. The evaporation rate increases with increasing interfacial mass fraction, and all simulations approach the value provided by Eq. (151) after a transient relaxation period of ~ 6 ms. Panel (b) provides the corresponding errors $E_{\dot{m}}$. At the highest interfacial mass fraction, $Y_\Gamma = 0.9$, the error reaches a minimum of $\sim 5\%$ for $t > 6$ ms. The $Y_\Gamma = 0.6$ and $Y_\Gamma = 0.3$ achieve smaller errors of $E_{\dot{m}} = 2.0\%$ and $E_{\dot{m}} = 0.7\%$, respectively.

5.4. Stationary evaporating droplet

A spherical droplet evaporating in quiescent gas achieves a pseudo steady-state when the far-field temperature and species mass fraction are held constant. This configuration has been used to verify previous evaporation methodologies [13,17].

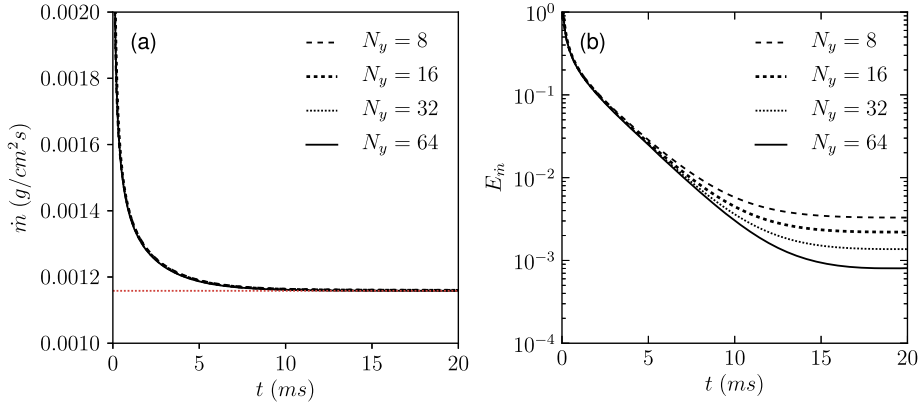


Fig. 10. Mesh sensitivity of evaporation rate $\dot{m}$ from the surface of a planar film with interfacial vapor mass fraction $Y_\Gamma = 0.1$. Panel (a) illustrates the transient relaxation to steady-state for mesh resolutions of $N_y = 8, 16, 32$, and 64 (cases 1-4, respectively), where the dotted red line indicates the analytical solution. Panel (b) provides the corresponding evaporation rate error $E_{\dot{m}}$, which decreases as the solution approaches steady-state.

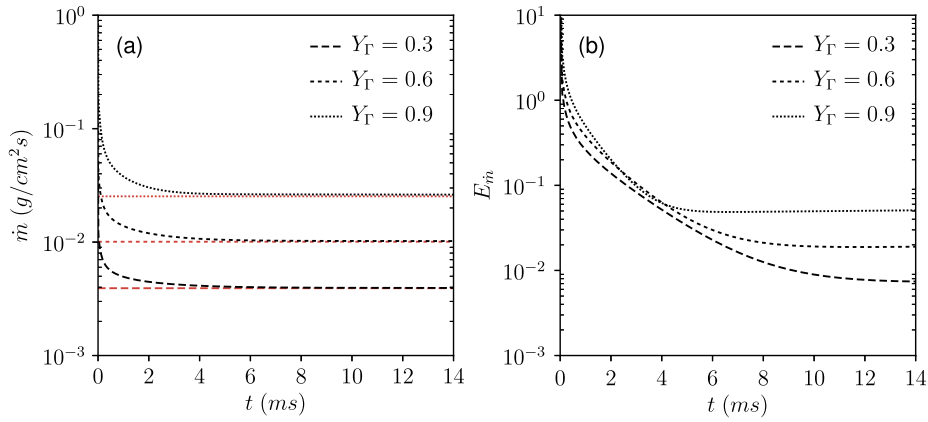


Fig. 11. Evaporation from the surface of a one-dimensional film on an $N_y = 16$ mesh for interface vapor mass fractions of $Y_\Gamma = 0.3$ (Case 5), $Y_\Gamma = 0.6$ (Case 6), and $Y_\Gamma = 0.9$ (Case 7). Panel (a): temporal relaxation of the evaporation rate $\dot{m}$ to the steady condition. Panel (b): temporal relaxation of the normalized evaporation rate error $E_{\dot{m}}$.

Problem setup The evaporation rate of a spherical droplet in quiescent gas is given by

$$\dot{m} = \frac{2}{d} \rho_g \mathcal{D} \ln \left(1 + \frac{Y_\Gamma - Y_\infty}{1 - Y_\Gamma} \right), \quad (157)$$

where d is the droplet diameter at time t , Y_Γ is the vapor mass fraction at the interface, Y_∞ is the far-field vapor mass fraction, ρ_g is the gas density, and $\mathcal{D}$ is the coefficient of mass diffusion [17,13]. The time required to completely evaporate the droplet is given by

$$\tau = \frac{\rho_l d_o^2}{8 \rho_g \mathcal{D} \ln(1 + B)} \quad (158)$$

where

$$B = \frac{Y_\Gamma - Y_\infty}{1 - Y_\Gamma}. \quad (159)$$

The diameter ratio at time t relative to the initial time t_o is then given as a function of τ by

$$\left(\frac{d}{d_o} \right)^2 = 1 - \frac{t}{\tau}. \quad (160)$$

Equation (160) is known colloquially as the d^2 law. Verification is performed by comparing the simulated droplet diameter to Eq. (160).

Table 5
Fluid properties for evaporating spherical droplet verification.

Property	Iso-octane	Air
k [erg/s cm K]	1.69×10^4	6.316×10^3
μ [g/cm s]	5.8×10^{-3}	1.871×10^{-4}
$\mathcal{D}$ [cm ² /s]	0.035	0.035

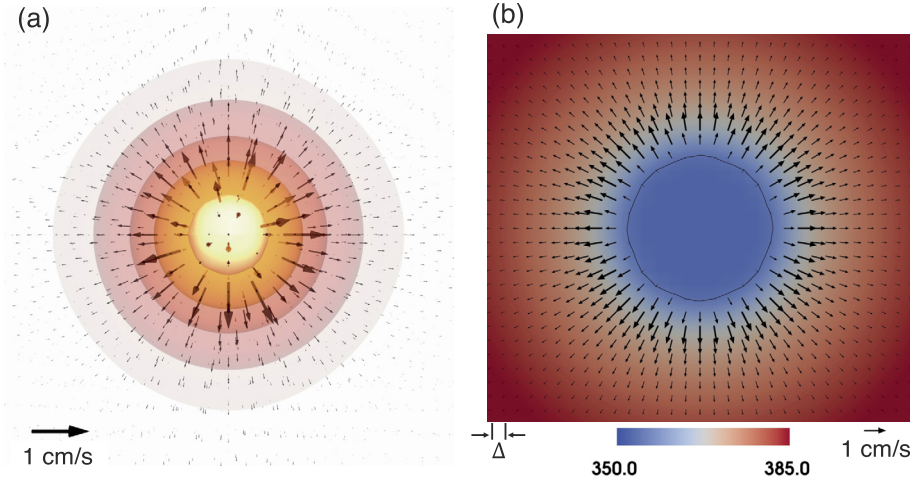


Fig. 12. Evaporation of a stationary droplet simulated on an $N_x = 64$ mesh and visualized at $t = 1.2$ ms: (a) vapor mass fraction (translucent iso-surfaces), velocity vectors, and droplet surface (opaque white sphere); progressing from outside in, the iso-surfaces correspond to $Y = 0.06$, $Y = 0.075$, $Y = 0.1$, and $Y = 0.15$; (b) Temperature (in Kelvin) and velocity vectors, visualized in the near-droplet region on a plane bisecting the droplet. The mesh spacing Δ appears in the lower-left corner.

Numerical specification A single iso-octane droplet with initial radius $d_0 = 100 \mu\text{m}$ is suspended, surrounded by air, in a domain with side length $L = 800 \mu\text{m}$. The analytical description of this configuration has been extensively covered in prior work [17,13], and is not repeated in detail. The far-field pressure p_∞ , the far-field temperature T_∞ , and the fluid properties provide closure for the analytical solution. This assessment considers a single pressure, $p_\infty = 900$ kPa, and four temperatures: $T_\infty = 300, 350, 400$, and 450 K. Each far-field temperature corresponds to a different interface temperature, gas temperature profile, and vapor mass fraction profile. These analytical profiles for temperature and mass fraction, along with the far-field pressure, are used to initialize the gas phase properties. The liquid temperature is initialized uniformly equal to the interface temperature. Equations of state, specific internal energies, and enthalpies of evaporation are defined by the REFROP thermodynamics library [42]. To simplify use of Eq. (160), the species diffusion coefficient is fixed to a value of $\mathcal{D} = 0.035$ cm²/s. The viscosities and thermal conductivities of the liquid and gas phases are also considered constant, and are set equal to the values provided in Table 5.

Each of the four temperature configurations is simulated on a uniform mesh of $N_x = N_y = N_z = 64$ mesh spacings. The $T_\infty = 450$ K case is additionally considered at mesh resolutions of $N_x = 32$ and $N_x = 128$ to allow for an assessment of mesh sensitivity. Neumann conditions are applied at the domain boundaries, and mass exits such that $p = p_\infty$ at the domain boundary.

The analytical solution, Eq. (160), makes a number of assumptions, including the presence of a uniform density field, a fixed interface temperature, and a fixed interfacial vapor mass fraction. In an effort to exercise the complete numerical implementation, none of the constraints of the analytical solution are imposed on the simulation: the interfacial vapor mass fraction, interfacial temperature, and velocity field are not constrained to fixed values, but evolve as a function of the model. In order to generate a pseudo-analytical solution for comparison, the quantities present in Eq. (158) must be appropriately selected. The gas density ρ_G is set equal to the initial film density

$$\rho_G = \frac{\rho_{g,\infty} + \rho_{g,\Gamma}}{2}, \quad (161)$$

and the interfacial vapor mass fraction Y_Γ is defined by the average interfacial temperature and pressure derived from the simulation at time $t = 0.004$ ms, rather than fixing the interfacial vapor mass fraction to a constant.

Results Fig. 12 provides a snapshot of the solution field for the $T_\infty = 400$ K drop at time $t = 0.5$ ms. Panel (a) shows iso-surfaces of vapor mass fraction, velocity vectors, and the droplet surface (the interior white sphere). Evaporation induces flow away from the interface, with the velocity magnitude decreasing in the radial direction. Like the velocity magnitude, the

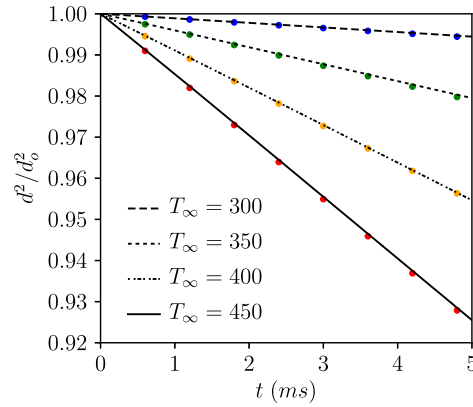


Fig. 13. Temporal reduction in d^2/d_0^2 for evaporating droplets suspended in air for far-field temperatures of $T_\infty = 300$ K, 350 K, 400 K, and 450 K. Corresponding analytical values of d^2/d_0^2 appear as blue, green, yellow, and red dots, in order of increasing far-field temperature.

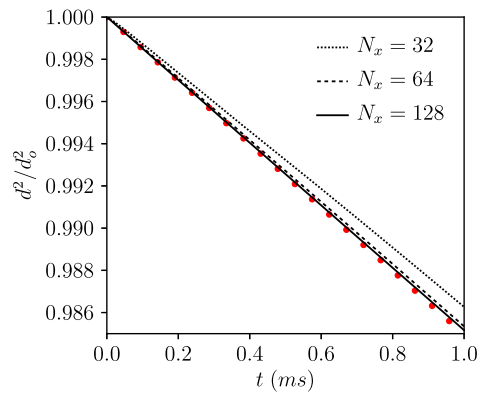


Fig. 14. Sensitivity of the droplet diameter d^2/d_0^2 to mesh resolution for $T_\infty = 450$ K, considered at resolutions of $N_x = 32$, 64, and 128 mesh spacings. The analytical value of d^2/d_0^2 is indicated by red dots.

maximum vapor mass fraction appears at the interface. In Panel (a), the innermost iso-surface, shown in orange, corresponds to a vapor mass fraction of $Y = 0.15$. Vapor mass fraction decreases rapidly in the radial direction, with the outermost iso-surface corresponding to $Y = 0.06$.

Fig. 12 panel (b) shows the near-droplet temperature field on a bisecting plane. The liquid/gas interface appears as a black contour, and velocity vectors illustrate the Stefan flow. The velocity is effectively zero in the liquid phase due to the large liquid-to-gas density ratio. Similarly, the liquid temperature is nearly constant, while the gas temperature varies strongly as a function of distance from the interface. Despite the highly coupled nature of the simulation and the relatively coarse mesh resolution, the flow field appears free from artifacts.

The normalized droplet diameter d^2/d_0^2 appears as a function of time in Fig. 13 for each of the far-field temperatures under consideration. As the far-field temperature increases from $T_\infty = 300$ K to $T_\infty = 450$ K, the droplet diameter decreases more rapidly in time. The droplet diameter defined by Eq. (160) appears in Fig. 13 as blue, green, yellow, and red circles, corresponding to $T_\infty = 300$, 350, 400, and 450 K, respectively. The simulated diameter reduction trails the value prescribed by Eq. (160) for all temperatures considered, suggesting that the $N_x = 64$ mesh resolution mildly under-predicts the evaporation rate.

Sensitivity of the $T_\infty = 450$ K evaporation dynamic to mesh resolution is examined in Fig. 14. The analytical solution is denoted by red dots. The low resolution $N_x = 32$ mesh discretizes the droplet by only four mesh spacings, resulting in an under-prediction of the evaporation rate. Error decreases significantly for the increased resolution of $N_x = 64$, and again with the additional increase to $N_x = 128$.

5.5. Evaporating droplet in translation

The analytical solution for an evaporating stationary droplet considered in Section 5.4 also applies to a liquid/gas system in uniform translation. This configuration further stresses the methodology by introducing numerical errors associated with advection.

Numerical specification The configuration used to examine the d^2 law from Section 5.4 is essentially repeated, with a few modifications required to implement translation of the liquid and gas relative to the computational mesh. The $T_\infty = 450$ K temperature is considered for mesh resolutions of $N_x = 32, 48, 64$, and 96 . The initial condition for each case is prescribed as outlined in Section 5.4, but a uniform translational velocity is applied to both phases and all three velocity components: $u_x = u_y = u_z = 25$ cm/s. Periodicity is imposed at all domain boundaries, rather than the outflow condition considered in Section 5.4.

Verification of the evaporation rate is again performed by comparison to Eq. (160). The simulation error is quantified via the diameter error E_{d^2} ,

$$E_{d^2} = \frac{d_s^2 - d^2}{d_o^2 - d^2}, \quad (162)$$

where d_s is the simulated diameter and d is the diameter provided by Eq. (160). Additionally, verification of global conservation is performed by considering temporal changes in mass, momentum, and energy. Conservation errors are nondimensionalized according to

$$E_c^*(t) = \frac{\int_\infty Q(t) dV}{\int_\infty Q(t_o) dV}, \quad (163)$$

where E_c^* is the dimensionless conservation error, $Q(t)$ is the conserved quantity at time t , and $Q(t_o)$ is the conserved quantity at time t_o . Conservation errors are considered for two different configurations of solution tolerance variable, $tol = 10^{-8}$ and $tol = 10^{-12}$. The solution tolerance variable controls two components of the solution procedure: it defines the smallest realizable volume fraction (volume fractions smaller than tol are truncated), and it determines the convergence tolerance of the interfacial system Eq. (93). Finally, we consider the frequency with which a tolerance-satisfying solution is not found for the interfacial transport and phase change system, Eq. (93). Failure to converge activates the procedure outlined in Appendix A. This frequency is quantified with parameter $\mathcal{F}_{A_\Gamma}$

$$\mathcal{F}_{A_\Gamma} = \sum_n \left[\frac{\sum_i A_{\Gamma,i} \delta_i}{\sum_i A_{\Gamma,i}} \right], \quad (164)$$

where n is the time level, i is the computational cell index, $A_{\Gamma,i}$ is the interfacial area contained in cell i , and δ_i is the delta function defined by

$$\delta_i = \begin{cases} 0 & \text{if Eq. (93) converged} \\ 1 & \text{else} \end{cases}. \quad (165)$$

Equation (164) is therefore a measure of the total interfacial area associated with convergence failure, divided by the total interfacial area.

Results The ability of the method to capture evaporation in the translating droplet system is assessed in Fig. 15. Panel (a) illustrates the temporal reduction in diameter for each of the four mesh resolutions considered, with the analytical diameter indicated by red circles. Each of the mesh resolutions considered over-predicts the rate of diameter reduction, with increasing over prediction corresponding to decreasing mesh resolution. This behavior is consistent with the introduction of diffusion error associated with the system translating through the computational mesh. These errors are quantified in Panel (b) via the diameter error E_{d^2} . After a brief transient related to the initial condition, the errors stabilize for each resolution considered. Error magnitudes decrease with increasing mesh resolution.

Fig. 16 provides conservation errors for mass (solid red), momentum (dotted green), and energy (dashed blue) for the $N_x = 48$ simulation. Results are shown for two values of the solution tolerance variable, $tol = 10^{-8}$ (thick lines) and $tol = 10^{-12}$ (thin lines). Conservation errors are $\mathcal{O}(10^{-8})$ for $tol = 10^{-8}$, and errors decrease to $\mathcal{O}(10^{-11})$ for $tol = 10^{-12}$. Conservation errors are therefore controlled by the convergence tolerances applied to the interface transport systems and the moment of fluid reconstruction. The run time considered allows the droplet to traverse the full domain and pass through the periodic boundary. Over the course of the simulation, nearly 8% of the liquid mass evaporates into the gas phase. For the low tolerance simulation, $tol = 10^{-12}$, a small fraction of total interfacial area fails to converge, with a convergence failure metric of $\mathcal{F}_{A_\Gamma} = 4.353 \times 10^{-7}$. Increasing the convergence tolerance has little impact on failure occurrence, with $\mathcal{F}_{A_\Gamma} = 1.496 \times 10^{-7}$ for the larger tolerance of $tol = 10^{-8}$.

5.6. Evaporating droplet in crossflow

Renksizbulut et al. [43] present a correlation for the evaporation rate of a droplet suspended in a high-temperature crossflow. This correlation has previously been used to validate the performance of computational evaporation models [9]. Comparison to this correlation provides validation and a final assessment of the present model. In addition to the accuracy assessment, we consider the convergence failure metric, and the computational expense of the method as a function of mesh size.

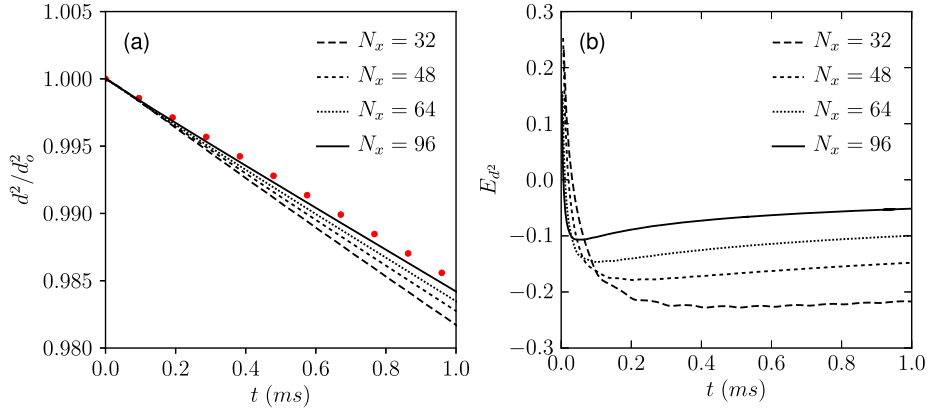


Fig. 15. Temporal reduction in diameter of iso-octane droplets in a translating liquid/gas system: (a) normalized diameter d^2/d_o^2 for mesh resolutions $N_x = 32, 48, 64$, and 96 , with the analytical solution represented by red dots; (b) diameter error E_{d^2} .

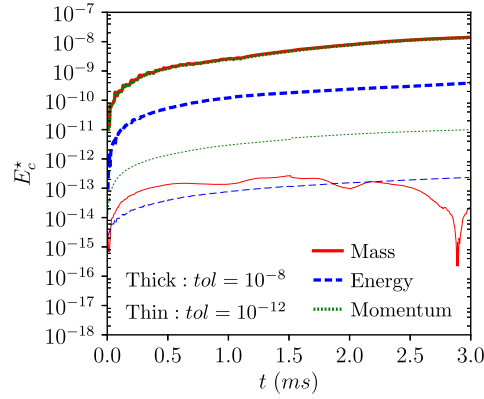


Fig. 16. Conservation error for mass, momentum, and energy for the $N_x = 48$ mesh in the uniform advection test problem. Thick lines correspond to a solution tolerance variable of $tol = 10^{-8}$, and thin lines correspond to $tol = 10^{-12}$.

5.6.1. Problem setup

The correlation of Renksizbulut et al. [43] relates the droplet evaporation rate to the Reynolds number, the Sherwood number Sh , the Schmidt number Sc , and the mass transfer number B_M , which are defined as follows:

$$Re_d = \frac{\rho_\infty U_\infty d}{\mu_{film}}, \quad (166)$$

where ρ_∞ is the free stream gas density, U_∞ is the free stream velocity, μ_{film} is the gas phase viscosity evaluated at the film temperature, and d is the droplet diameter;

$$Sh_{film} = \frac{dh_m}{\mathcal{D}_{film}}, \quad (167)$$

where h_m is the mass transfer coefficient, and $\mathcal{D}_{film}$ is the mass diffusion coefficient evaluated at the film temperature;

$$Sc_{film} = \frac{\mu_{film}}{\rho_{film} \mathcal{D}_{film}}, \quad (168)$$

where ρ_{film} is the gas phase density evaluated at the film temperature, and

$$B_M = \frac{Y_\Gamma - Y_\infty}{1 - Y_\Gamma}, \quad (169)$$

where Y_∞ is the free stream vapor mass fraction. Using these dimensionless parameters, Renksizbulut et al. [43] provide the following correlation:

$$Sh_{film} (1 + B_M)^{0.7} = 2 + 0.87 Re_d^{1/2} Sc_{film}^{1/3}. \quad (170)$$

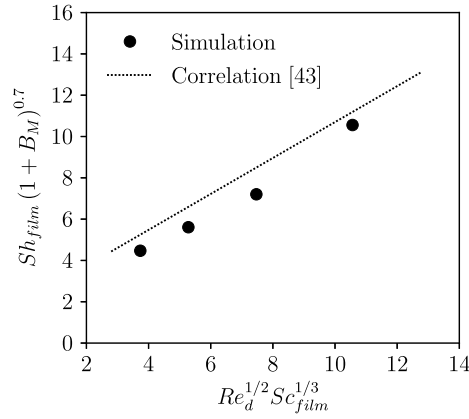


Fig. 17. Simulation results for iso-octane droplets in high temperature crossflow (circles) compared to the empirical correlation of Renksizbulut et al. [43] (dotted line).

This correlation has been designed for the high evaporation rate regime, where the thickening of the species and thermal boundary layers at increasing mass transfer rates has an insulating effect, and reduces the effectiveness of phase change [43,9].

5.6.2. Numerical specification

A $d = 200 \mu\text{m}$ iso-octane droplet is suspended in a crossflow of hot, dry air. The simulation domain is a box with side lengths $L_x = 1500 \mu\text{m}$ in the x -direction, $L_y = 1200 \mu\text{m}$ in the y -direction, and $L_z = 1200 \mu\text{m}$ in the z -direction. Air enters from the left boundary at free stream velocity U_∞ and temperature $T_\infty = 573 \text{ K}$, and exits the right boundary at pressure $p_\infty = 300 \text{ kPa}$. Free-stream velocities of $U_\infty = 62.5, 125, 250$, and 500 cm/s are considered to examine the effects of crossflow velocity on the rate of phase change. The droplet is initially stationary and at temperature $T_d = 350 \text{ K}$. The liquid/gas system is defined by a constant surface tension coefficient of $\sigma = 12.0 \text{ dynes/cm}$.

Block-structured automatic mesh refinement, facilitated by the AMReX library, is used to reduce the computational cost of this test. A base mesh of $N_x = 80$ by $N_y = N_z = 64$ is refined near the droplet by two additional mesh levels, resulting in a near-droplet mesh spacing of $\Delta_x = 9.375 \mu\text{m}$.

In order to represent the dynamics of a high-temperature evaporation scenario, this validation case considers variable transport properties in the gas phase (the liquid phase uses the same properties as used for the iso-octane verification tests). The coefficient of mass diffusion is defined in cgs units by Chapman-Enskog theory as a function of temperature and pressure according to

$$D = 16.1426 \frac{T^{3/2}}{p}. \quad (171)$$

The dynamic viscosity and thermal conductivity of dry air are defined as a function of temperature and density according to Kadoya et al. [44]. The thermal conductivity in the gas phase is defined as the mass-averaged value of dry air and iso-octane vapor, the latter assumed to be a constant of $k = 8 \times 10^4 \text{ erg/s} \cdot \text{cm}^2 \cdot \text{K}$. The dynamic viscosity of the mixture is assumed to equal that of dry air.

Results Comparison of the model results to the empirical correlation is presented in Fig. 17. Strong agreement is achieved across the tested range of correlation parameter $Re_d^{1/2} Sc_{film}^{1/3}$, suggesting the method is capable of realizing the suppression of phase change corresponding to the thickening of species and thermal boundary layers [43,9].

Readdressing the convergence failure metric for this more realistic validation problem, we find $\mathcal{F}_{Ar} = 2.58 \times 10^{-6}$ for the $U_\infty = 250 \text{ cm/s}$ simulation; the interfacial system is therefore well-defined with a tolerance-satisfying solution in nearly all scenarios encountered in the simulation considered.

The computational cost of the method is considered for the $U_\infty = 250 \text{ cm/s}$ droplet in Fig. 18 for six different computational meshes. Each mesh is defined by a base resolution and two additional levels of near-interface refinement via the AMReX library. The total number of control volumes, defined as the cell count, appears as the x-axis in Fig. 18. The smallest cell count shown in Fig. 18 corresponds to the mesh used to produce the results in Fig. 17. The computational cost of the method, measured in core seconds per iteration, is defined as the product of wall time required to advance the simulation from time level n to time level $n + 1$, and the number of MPI ranks used by the computation. All simulations were run on the Sandia Attaway cluster with 36 MPI ranks per compute node. The two smallest cell counts were run on $N_{nodes} = 4$ compute nodes, the three intermediate cell counts were run on $N_{nodes} = 16$ compute nodes, and the largest case was run on $N_{nodes} = 32$ compute nodes. As demonstrated by Fig. 18, roughly linear scaling is observed across the range of mesh sizes and node configurations considered.

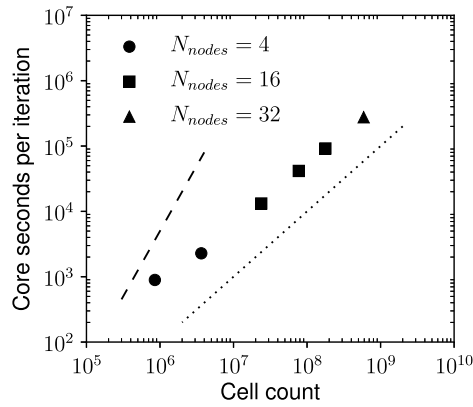


Fig. 18. Computational cost per iteration for the droplet in crossflow validation problem. The dotted and dashed lines illustrate first-order and second-order scaling with respect to cell count, respectively.

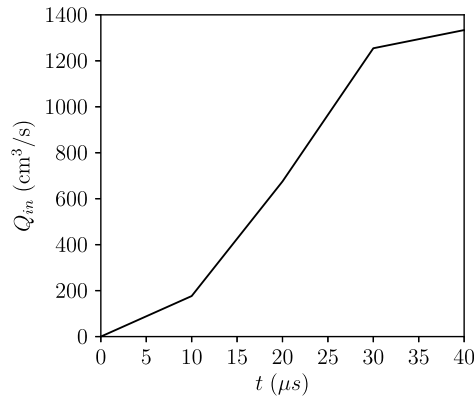


Fig. 19. Time-dependent volumetric flow rate Q_{in} of iso-octane.

5.7. High pressure evaporating spray

As a final capability demonstration, we consider a high pressure iso-octane spray, emulating a gasoline direct injection system.

5.7.1. Problem setup

The simulated injector is a sharp-edged orifice with diameter $d = 170 \mu\text{m}$ and an orifice length of $L = 250 \mu\text{m}$. The initial conditions for temperature and pressure are defined by ECN “Spray-G”. Measurements of mass flow rate through the Spray-G injector have been provided by the ECN [45], and the early opening transient is shown in Fig. 19. This volumetric flow rate is applied in the form of a time-dependent inlet velocity for the iso-octane upstream of the simulated injector. The iso-octane and air properties are defined identically to those used for the validation presented in Section 5.6. Adaptive mesh refinement near the interface is managed by the AMReX library, utilizing four levels of mesh refinement to achieve a near-interface mesh spacing of $\Delta_x = 1.76 \mu\text{m}$.

Results Fig. 20 illustrates the early evolution of the iso-octane spray via five instantaneous snapshots in panels (a)-(e). The spray issues from the injector in the downward direction, with time increasing from left to right. The liquid/gas interface is shown in opaque yellow, and the injector body appears as a translucent gray cylinder. The 10% mass fraction of iso-octane vapor is shown in translucent orange. As the spray evolves in time (from left to right), the volumetric flow rate increases, and the initially laminar flow transitions into a turbulent spray. The iso-octane atomizes and evaporates, and the simulation remains well behaved, despite the highly nonlinear and coupled interface dynamics. The total mass of iso-octane vapor contained in the system, m_γ , is shown as a function of time in panel (f). For the first 20 μs of the injection event, the total vapor mass changes negligibly from the initial condition. After 20 μs , the total vapor mass increases rapidly.

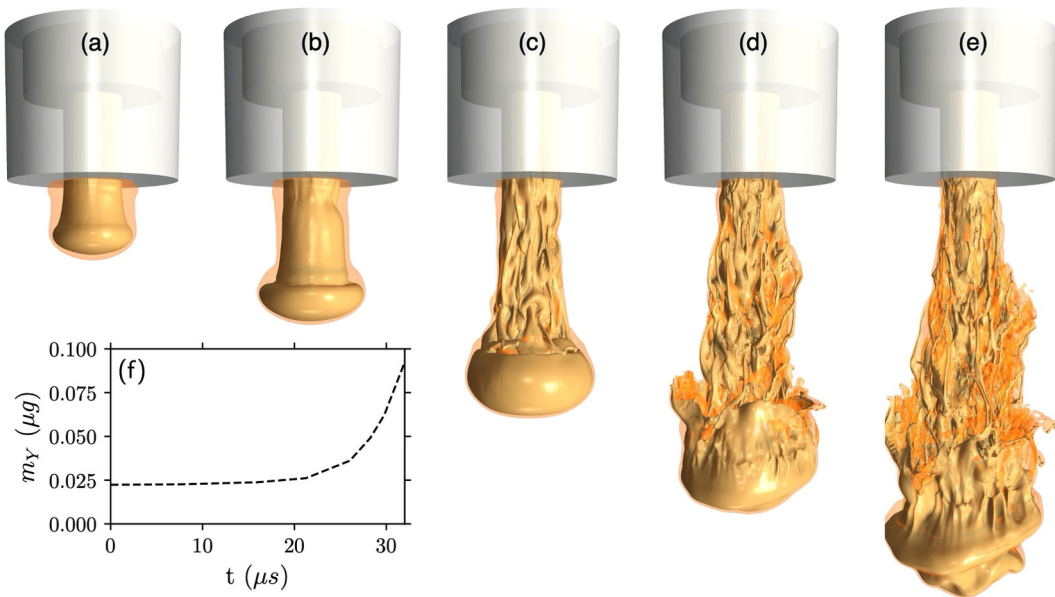


Fig. 20. Early development of an evaporating iso-octane spray exiting a $D = 170 \mu m$ sharp-edged orifice, shown in panels (a)–(e), along with the evolution of total iso-octane vapor mass, m_v , shown in (f). In (a)–(e), the 10% vapor mass fraction iso-surface appears translucent orange, and the liquid-gas interface appears opaque yellow. Temporal development is illustrated, progressing from left to right, at times: (a) $t = 15 \mu s$; (b) $t = 20 \mu s$; (c) $t = 25 \mu s$; (d) $t = 27 \mu s$; (e) $t = 32 \mu s$.

6. Summary and conclusions

We have presented a new framework for the modeling and simulation of phase transition between a liquid and a mixture of vapor and non-condensable gas separated by a sharp interface. Motivated by the need for predictive methods with sufficient robustness and versatility for problems of engineering interest, the framework is characterized by: (1) discrete consistency between phase change, mass transport, and energy transport at the interface, achieved efficiently by an implicit system localized at each interfacial cell; (2) conservation of mass, momentum, and energy; (3) accurate definition of the normal velocity discontinuity at the interface, derived from the single-fluid velocity and the rate of phase change; and (4) applicability to compressible and incompressible regimes by virtue of the asymptotically-preserving property of the underlying pressure projection scheme. The method is general with respect to interface topology, does not rely on global pressure or temperature assumptions to calculate phase change, and mitigates numerical artifacts near the interface typically associated with VOF methods. Although the framework has been presented and implemented in the context of the moment of fluid method, extension to other geometric VOF methods is trivial. Similarly, this paper focuses on phase change below the saturation temperature and in the presence of a non-condensable species, but the extension to boiling is straight forward, and a focus of future efforts (with preliminary results presented by Wenzel and Arienti [38]).

Verification exercises have demonstrated the ability of the method to capture single phase shock waves with accuracy comparable to other asymptotically preserving schemes [37,36], and to provide consistent, conservative, and convergent descriptions of interfacial heat transfer and phase change. Verification of evaporation has been performed in planar, stationary droplet, and translating droplet configurations as a function of mesh resolution and evaporation rate. In each case, robustness and accuracy are observed at the low numerical resolutions expected in engineering analysis, where level-set-based phase change methodologies are prone to conservation errors. Using the AMReX library for automatic mesh refinement, validation of the methodology has been presented for iso-octane droplets in high temperature crossflow, demonstrating strong agreement with the empirical correlation of Renksizbulut et al. [43].

The local nature of the mixed-time integration scheme developed for the coupling of phase change and diffusion allows for affordable and efficient application to problems of engineering interest on highly parallel HPC systems, demonstrated by linear scaling of total computational cost with simulation size for the droplet in crossflow problem. This capability has been practically demonstrated by simulating a high pressure iso-octane spray. The iso-octane spray accounts for real fluid thermodynamics, and enforces consistent transport of mass, energy, and phase change as a function of variable, realistic fluid properties. The presented methodology appears to be the first capable of performing predictive simulations of this type, without simplification of the fluid properties. Having demonstrated verification, validation, and applicability to a realistic problem configuration, the presented method is a promising new tool for studying phase change systems.

CRediT authorship contribution statement

Both authors contributed to all aspects of the work presented in this article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Nearly homogeneous cells

Nearly homogeneous cells, or cells containing a small volume fraction of liquid or gas, require special consideration for both inter- and intra-phase diffusion. Cells undergoing evaporation with very small volume fractions of liquid, and cells undergoing condensation with very small volume fractions of gas, can fail to satisfy the implicit system defined by Eq. (93). This results from considering the phase change dynamic within a single computational cell: sufficient mass of both phases must be present to satisfy the rate of phase change over the time step Δt . In this section, we provide a number of alternative modeling approaches that preserve conservation for these off-design cases. The parameter ϵ is used to signify values of the VOF that are considered small. In the implementation of the method presented in this work, an attempt is always made to solve the implicit system, Eq. (93). Only if convergence fails does the methodology make use of the techniques in this section.

Thermal transport is considered from a lumped perspective when the gas or liquid VOF becomes small: $\phi_g^a < \epsilon$ or $\phi_l^a < \epsilon$. The two phases are assumed to be in thermal equilibrium at the mixture temperature T_m , satisfying

$$\rho_g^a e_g^a \phi_g^a + \rho_l^a e_l^a \phi_l^a = \rho_g^a e(T_m, Y^a)_g \phi_g^a + \rho_l^a e(T_m, \rho_l^a)_l \phi_l^a, \quad (\text{A.1})$$

where e is the specific internal energy. Equation (A.1) is solved for the mixture temperature T_m with a bisection method. After enforcing equilibrium between the phases, the external thermal fluxes are imposed on the cell in a lumped approach according to

$$e_l^* = e(T_m, \rho_l^a)_l + \frac{D_l^a + D_g^a}{\rho^{n+1}} \quad (\text{A.2})$$

and

$$e_g^* = e(T_m, Y^a)_g + \frac{D_l^a + D_g^a}{\rho^{n+1}}, \quad (\text{A.3})$$

where e_l^* and e_g^* are intermediate energies, and ρ^{n+1} is the volume-averaged density,

$$\rho^{n+1} = \rho_l^a \phi_l^a + \rho_g^a \phi_g^a. \quad (\text{A.4})$$

Next, we account for phase change and external flux of vapor, D_v^a . Treatment of evaporation depends on which phase is sparse. When the gas volume fraction is small, $\phi_g^a < \epsilon$, the gas phase is assumed to be at the saturation condition defined by the liquid temperature, $Y^d = Y_\Gamma(T_l^d, \rho_g^a)$

$$\rho_g^d Y_\Gamma(T_l^d, \rho_g^a) \phi_g^d = \rho_g^a Y^a \phi_g^a + \dot{m}^d \bar{A}_\Gamma^a \Delta t + D_v^a \Delta t. \quad (\text{A.5})$$

Assuming the change in VOF is negligibly small over the time step Δt , we can express the time-advanced density ρ_g^d as a function of the evaporation rate $\dot{m}^d$ and the advected density ρ_g^a :

$$\rho_g^d = \rho_g^a + \dot{m}^d \bar{A}_\Gamma^a \Delta t. \quad (\text{A.6})$$

Substituting Eq. (A.6) into Eq. (A.5) returns the evaporation rate for cells with $\phi_g^a < \epsilon$:

$$\dot{m}^d = \frac{\phi_g^a \rho_g^a (Y^a - Y_\Gamma) + D_v^a \Delta t}{\bar{A}_\Gamma \Delta t (Y_\Gamma - 1)}. \quad (\text{A.7})$$

For cells with only a small amount of liquid, $\phi_l^a < \epsilon$, we simply compute the evaporation rate explicitly, considering the evaluation of the derivative will occur over a distance of $\sim \Delta_x/2$. In this case, the evaporation rate is given by

$$\dot{m}^d = \min \left[\frac{0.75 \phi_l^a \rho_l^a}{\bar{A}_\Gamma \Delta t}, \rho_g^d \mathcal{D} \left(Y_\Gamma(T_l^a, \rho_g^a) - Y_g^d \right) (1 - Y_\Gamma(T_l^a, \rho_g^a))^{-1} \Delta_{g,\Gamma}^{-1} \right], \quad (\text{A.8})$$

where a limiter has been implemented such that no more than 75% of the liquid present in the cell evaporates over a single time step. Having established the evaporation rate, the remainder of the procedure is the same for cells with $\phi_l^a < \epsilon$ and cells with $\phi_g^a < \epsilon$.

The liquid VOF is updated according to

$$\phi_l^d = \left(\phi_l^a \rho_l^a - \dot{m}^d \bar{A}_\Gamma \Delta t \right) / \rho_l^d, \quad (\text{A.9})$$

which defines the time-advanced gaseous VOF

$$\phi_g^d = 1 - \phi_l^d. \quad (\text{A.10})$$

The gas density and vapor mass fraction are updated according to

$$\rho_g^d = \left(\rho_g^a \phi_g^a + \dot{m}^d \bar{A}_\Gamma \Delta t \right) / \phi_g^d, \quad (\text{A.11})$$

and

$$Y^d = \left(\rho_g^a Y^a \phi_g^a + \dot{m}^d \bar{A}_\Gamma \Delta t + D_v^a \Delta t \right) / \left(\rho_g^d \phi_g^d \right). \quad (\text{A.12})$$

The enthalpy of vaporization is finally imposed in the lumped framework to the liquid phase

$$E_l^d = E_l^\star - \frac{[h(T_m, \rho_g^a)] \dot{m}^d \bar{A}_\Gamma \Delta t}{\rho^{n+1}}, \quad (\text{A.13})$$

and the gas phase

$$E_g^d = E_g^\star - \frac{[h(T_m, \rho_g^a)] \dot{m}^d \bar{A}_\Gamma \Delta t}{\rho^{n+1}}. \quad (\text{A.14})$$

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